WEEK 1

Signals

Signals

A signal is a mathematical function used to represent information that varies with some independent variable.
For this unit, that independent variable is time.

Continuous time (CT)

$x(t) \quad t \in \mathbb{R}$

- $x(t)$: Entire Signal
- t: One signal value at particular time, t
- $\mathbb{R}$: Real Value

Discrete time (DT)

$x[n] \quad n \in \mathbb{Z}$

- $x[n]$: Entire Signal
- n: One signal value at particular time, n
- $\mathbb{Z}$: Integers

Digital Processors / Computers only understands numbers not continuous waves. Therefore **Sampling** is introduced to connect the two.

Suppose the original signal is $x(t)$, and we measure a singular value every T_s seconds.

$$x_s[0] = x(0)$$
$$x_s[1] = x(1\,T_s) \quad \text{After } T_s \text{ seconds}$$
$$x_s[1] = x(1\,T_s) \quad \text{After } T_s \text{ seconds}$$
$$\vdots$$

$$x_s[n] = x(n\,T_s)$$

$$\text{Sampling frequency } f_s = \frac{1}{T_s}$$

where T_s = Sampling period in Seconds / Sample
n = which sample
$n\,T_s$ = The actual CT time at which the sample was taken.

Signal / Function Manipulation

Transformations

Scaling [Vertical Stretch]

$y(t) = \alpha x(t)$

Add [Superposition]

$f(x) = x(t)$

$+ \quad g(x) = y(t)$

$= x(t) + y(t)$

Time Shift [Horizontal Translation]

Shift RIGHT by α

$y(t) = x(t - \alpha)$

Shift LEFT by α

$y(t) = x(t + \alpha)$

Time Dilation [Horizontal Stretch]

$y(t) = x(\alpha t)$

Take Note:

$\alpha < 0$ means reflection about the y-axis happens

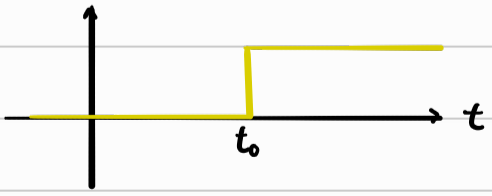
Building Block Signals

The course next introduces simple signals that are commonly used to build complicated signals.

① Unit Step [On / Off switch]

Continuous Time

$$u(t - t_0) = \begin{cases} 0 & \text{if } t < t_0 \\ 1 & \text{if } t > t_0 \end{cases}$$



Discrete Time

$$u(n - n_0) = \begin{cases} 0 & \text{if } n < n_0 \\ 1 & \text{if } n \geq n_0 \end{cases}$$

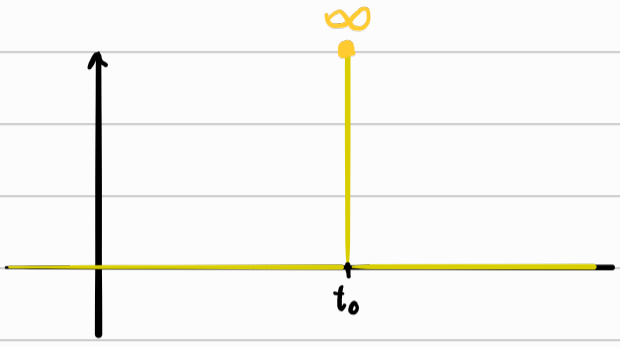
A pulse can be made as [(Turn on at a) - (Turn on at b)]

CT : $u(t - a) - u(t - b)$ OR DT : $u[n - a] - u[n - b]$

② Unit Impulse [Isolate 1 time location]

Continuous [Dirac Delta]

$$\delta(t - t_0) = \begin{cases} \infty & \text{if } t = t_0 \\ 0 & \text{if } t \neq t_0 \end{cases}$$



Properties : $\int_{t_0-\varepsilon}^{t_0+\varepsilon} \delta(t - t_0)\, dt = 1$

Sifting Property

$$\int_{-\infty}^{\infty} x(t)\, \delta(t - t_0)\, dt = x(t_0)$$

Discrete [Kronecker Delta]

$$\delta[n - n_0] = \begin{cases} 1 & \text{if } n = n_0 \\ 0 & \text{if } n \neq n_0 \end{cases}$$

Sifting Property

$$\sum_{-\infty}^{\infty} x[n]\, \delta[n - n_0]\, dn = x[n_0]$$

③ Complex Exponentials [Frequency Building Block]

Using Euler's identity,

$$e^{j\theta} = \cos\theta + j\sin\theta$$

CT unit complex exponential : $x(t) = e^{j(\omega t)}$

Angular Frequency
[ωt means as t marches forward, the angle, $\theta = \omega t$ grows faster]

Scaled & shifted ↓

$$e^{(\sigma + j\omega)t} = [e^{\sigma t}][e^{j\omega t}]$$

Growth / Decay — Rotation / Oscillation

Periodicity

This just tells us when the same behaviour repeats.

A CT is periodic if some $T > 0$ satisfies

$$x(t+T) = x(t) \quad \text{for all values } t$$

Fundamental Angular Frequency $\omega_0 = \frac{2\pi}{T_0}$

- ω_0: Angular Frequency [Rotational Speed]
- 2π: Angular Distance of one lap
- T_0: Time taken for one lap

A DT is periodic if some $N > 0$ satisfies

$$x[n+N] = x[n] \quad \text{for all integers } n$$

Fundamental Angular Frequency $\omega_0 = \frac{2\pi}{N_0}$

- ω_0: Angular Frequency [Rotational Speed]
- 2π: Angular Distance of one lap
- N_0: Time taken for one lap

The crucial difference btwn CT & DT is that ...

For CT, $e^{j(\omega t)} \Longrightarrow T = \frac{2\pi}{\omega}$

For DT, $e^{j(\omega n)} \Longrightarrow N = \frac{2\pi}{\omega}$ if N is an integer

↓ Mathematically

$$\omega N = (2\pi)m \quad \text{where } N, m \text{ are integers}$$

Hence a DT sinusoid / complex exponential is periodic only when

$\frac{\omega}{2\pi} \in \mathbb{Q}$ — Rational $\left[\frac{\text{Integer}}{\text{Integer}}\right]$

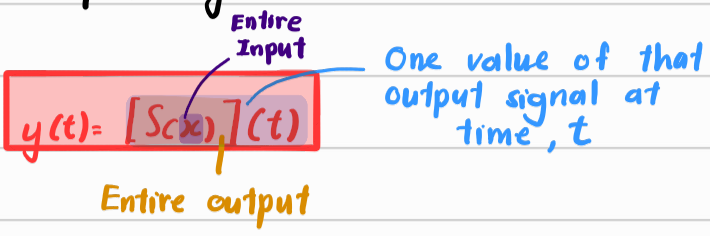
This is the reason "a sinusoid is periodic" is always safe in ordinary non-zero CT, but not automatically safe in DT.

WEEK 2

Systems

Systems

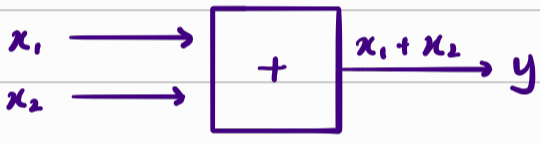
A system S is a rule that takes an entire input signal and produces an entire output signal



So $y = S(x)$ is effectively a function whose input itself is itself a function.

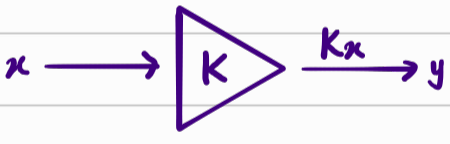
Building Block Systems

① Adder



$$y = x_1 + x_2 + \dots$$

② Gain



$$y = kx$$

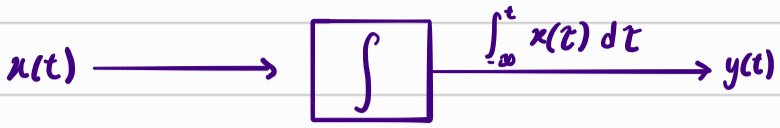
③ Delay

$$y(t) = x(t - \tau)$$

④ Full-wave rectifier

$$y = |x|$$

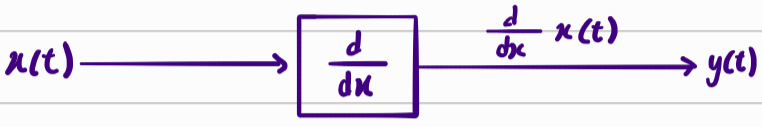
⑤ CT Integrator



Dummy Variable

$$y(t) = \int_{-\infty}^{t} x(\tau)\,d\tau$$

⑥ CT differentiator



$$y(t) = \frac{d}{dx}x(t)$$

Interconnected Systems

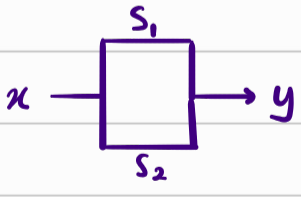
The Building Blocks can be interconnected to make a more complicated system.

① Cascade / Series

$$y = S_2[S_1(x)]$$

* Generally Order matters

② Parallel



$$y = S_1(x) + S_2(x)$$

③ Feedback

Part of the output is processed and returned to an earlier point.

A feedback eqn therefore contain "y" on both sides bcuz the current output participates in determining the system input signal.

System Properties

① Memoryless

Does output now use only input now?

Continuous-time

$y(t) = F(t, x(t))$

current inputs

Discrete-time

$y[n] = F(n, x[n])$

current inputs

② Causal

System where current output is only caused by current or/and past inputs. Future inputs cannot cause a change in output today.

CT

$y(t) = F(t, x(t), x(t-\tau))$

current inputs, past inputs

DT

$y[n] = F[n, x(n), x(n-N)]$

current inputs, past inputs

A memoryless system is a causal system
A causal system might not be a memoryless system

③ BIBO stability

BIBO means Bounded-Input Bounded-Output. Essentially if the input is bounded, a BIBO stable system output is also bounded.

finite values

$$\text{If } |x(t)| \leq M_x \implies |y(t)| \leq M_y$$

To prove BIBO ~ Show every bounded input gives bounded output
To disprove BIBO ~ Find one bounded input that gives an unbounded output

④ Time Invariance

A time invariant system behaves the same regardless of when the experiment is performed.

If

$$x(t) \xrightarrow{S} y(t)$$

then shifting the input by τ must give only the same shift in the output:

$$x(t-\tau) \xrightarrow{S} y(t-\tau)$$

A useful test is to see if the system & Delay commute : $S \cdot D_\tau = D_\tau \circ S$

Route 1 (System then delay) : $D_\tau \{ S(x(t)) \} = y(t-\tau)$

Route 2 (Delay then System) : $S\{ D_\tau (x(t)) \} = S\{x(t-\tau)\}$

$$\boxed{S\{x(t-\tau)\} = y(t-\tau)}$$

⑤ Linearity

A Linear System preserves linear combinations.

$$\boxed{S(\alpha x_1 + \beta x_2) = \alpha S(x_1) + \beta S(x_2)}$$

We can either use ↑ or test the separate properties of linearity

Homogeneity : $S(\alpha x) = \alpha S(x)$

Additivity : $S(x_1 + x_2) = S(x_1) + S(x_2)$

Prove Linearity : Proove both of these properties

Disprove Linearity : Disprove either one property

⑥ Linear Time Invariant (LTI)

This is the "Golden pair" bcuz it tells us what happens when an input is ...

1) Shifted
2) Scaled
3) Added to other inputs.

Difference & Differential Eqns

These are mathematical eqns to build real-world LTI systems.

① Non-recursive DT case

$$y[n] = b_0 x[n] + b_1 x[n-1] + \dots + b_M x[n-M]$$

The output is assembled from present/past input samples only

∴ This is naturally causal and LTI

② Recursive DT case

$$a_0 y[n] + a_1 y[n-1] + \dots + a_N y[n-N] = b_0 x[n] + b_1 x[n-1] + \dots + b_M x[n-M]$$

Now previous outputs feed into the new output.

∴ The eqn alone is not enough, we still need initial information.

[No pre-stored energy]

The course therefore introduces condition of initial rest.

If $x[n] = 0$ for all time before the starting time, $y[n] = 0$ too

∴ Under that condition, the causal recursive system behaves as an LTI system.

③ Recursive CT case

$$a_0 y + a_1 \frac{dy}{dt} + \dots + a_N \frac{d^N y}{dt^N} = b_0 x + b_1 \frac{dx}{dt} + \dots + b_M \frac{d^M x}{dt^M}$$

Instead of shift in time $(n-1)$, physical circuits use derivatives $\left(\frac{dy}{dt}\right)$

Again, recursive behaviour means initial conditions matter, for the system to be LTI, it also uses the condition of initial rest

The Biggest Takeaway

If the system only uses

1) fixed linear operations
2) fixed delays (or derivatives)
3) fixed gains

It forms an LTI system as long as it starts completely at rest

WEEK 3

Convolution

Representing Signals with Impulses

From Week 1, we have learned the kronecker delta, $\delta[n-k]$ as as a building block. Now we learn how to actually use this building block on DT signals

Trivially, a discrete signal is just a collection of vertical data values mapped to their respective time slots. Thus, it follows that a singular discrete data value $x[n]$ from the signal can be "sifted out" mathematically by running a "timeline scan" Σ and using a "selector" $\delta[n-k]$ to isolate only the target value $x[k] = x[n]$ of that particular time slot n

DT sifting formula →

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\,\delta[n-k]$$

- $x[n]$: Extracted target value
- $\sum$: Timeline Scanner
- $x[k]$: Value at current scan index
- $\delta[n-k]$: Selector

For example, lets say we want to extract the signal value at $n=1$, $x[1]$. Beuz of the "selector", $\delta[1-k]$ will only be "switched on [1]" when its arguement = 0, that is $k = 1$ ($k = n$), and all other time it will be "switched off [0]". So we get...

$$x[1] = \sum_{k=-\infty}^{\infty} x[k]\,\delta[1-k] = \{x[1]\,\underbrace{\delta[1-1]}_{1}\} + 0 + 0 \ldots = x[1]$$

∴ So instead of treating $x[n]$ as one complicated signal, we can think of it as:

$$x[n] = [\text{impulse at } k=0] + [\text{impulse at } k=1] + \ldots$$

where every impulse is...

1) Shifted to the time slot where the input sample occurs
2) Scaled to the actual sample value $x[k]$

Convolution

We just learned that using the DT sifting formula, an entire discrete-time signal can be represented as a sum of individual isolated signals.

Since our complete input can be broken down into shifted and scaled copies of the same building block $\delta[n]$, the natural question is

What does the system do to one basic unit impulse?

So we deliberately probe the system using

$$\delta[n]$$

which is a unit impulse applied at time $n = 0$. The system will then respond by producing an entire output signal, defined as impulse response, $h[n]$

$$\delta[n] \xrightarrow{S} h[n]$$

The complete response of the system to one unit impulse at $n = 0$

Assuming that we are dealing with an LTI system, once we know how the system responds to the fundamental impulse, $\delta[n]$, linearity & time invariance allows us to scale & shift the input impulse whilst knowing that the impulse response will also be scaled & shifted accordingly.

$$\delta[n] \xrightarrow{S} h[n]$$

$$\Downarrow \text{Shift}$$

$$\delta[n-k] \xrightarrow{S} h[n-k]$$

$$\Downarrow \text{Scale}$$

$$\underbrace{x[k]\,\delta[n-k]}_{\text{One isolated input sample}} \xrightarrow{S} \underbrace{x[k]\,h[n-k]}_{\text{Complete response caused by that sample}}$$

Important

One sample can affect several output times. An input sample existing only at time k does not mean its effect must also exist for only one time. That depends on the impulse response, $h[n]$.

For Example consider the 3-point avg system:

$$h[n] = \frac{1}{3}\delta[n] + \frac{1}{3}\delta[n-1] + \frac{1}{3}\delta[n-2]$$

with input sample

$$x[n] = \delta[n] - \delta[n-1] + 2\delta[n-2]$$

This $h[n]$ literally says, one input unit impulse at $n=0$ will create 3 separate output samples: $\frac{1}{3}$ at $n=0$, $\frac{1}{3}$ at $n=1$, $\frac{1}{3}$ at $n=2$.

$$\delta[n] \xrightarrow{S} h[n]$$

And cuz of LTI, an input sample of an impulse of height 2 at $n=2$, will then create $2h[n-2]$: $\frac{2}{3}$ at $n=2$, $\frac{2}{3}$ at $n=3$, $\frac{2}{3}$ at $n=4$.

$$2\delta[n-2] \xrightarrow{S} 2h[n-2]$$

Similarly for $-\delta[n-1]$: $-\frac{1}{3}$ at $n=1$, $-\frac{1}{3}$ at $n=2$, $-\frac{1}{3}$ at $n=3$

$$-\delta[n-1] \xrightarrow{S} -h[n-1]$$

Input Component	Complete response	Non-zero output times
$\delta[n]$ $x[0]$	$h[n]$	0, 1, 2
$-\delta[n-1]$ $x[1]$	$-h[n-1]$	1, 2, 3
$2\delta[n-2]$ $x[2]$	$2h[n-2]$	2, 3, 4

So lets say I want to find the output at $n=2$, $y[2]$

$$y[2] = \overset{x[0]}{1}\cdot h[2] + \overset{x[1]}{(-1)}\cdot h[2-1] + \overset{x[2]}{2}\cdot h[2-2]$$

$$y[2] = 1\cdot h[2] + (-1)\cdot h[1] + 2\cdot h[0]$$

$$= \left[\frac{1}{3}\right] + \left[-\frac{1}{3}\right] + \left[\frac{2}{3}\right]$$

$$y[2] = \frac{4}{3} \#$$

Note that $y[2]$ doesn't only depend on $x[2]$. Yes, $x[2]$ happens exactly at $n=2$, doesn't mean the previous samples $x[0]$, $x[1]$ doesn't contribute to the output at time $n=2$.

Taking a closer look at each term, at $n=2$ ...

$x[0] \cdot h[2]$: What is the contribution of $x[0]$ after 2 time units.

$x[1] \cdot h[2-1]$: What is the contribution of $x[1]$ after 1 time unit.

$x[2] \cdot h[2-2]$: What is the contribution of $x[2]$ at the current $n=2$

So essentially, $h[n-k]$ means

Take the impulse response produced by an impulse that occured at time "k", then look at the value of the response at output time "n".

∴ Generalising the reasoning,

every input sample

$$x[k]\,\delta[n-k]$$

produces its own complete response

$$x[k]\,h[n-k]$$

The final output is the sum of all these shifted/scaled response signals

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]\,h[n-k]$$

- $y[n]$: Extracted target output
- Target time n
- k: Time Scanner
- $x[k]$: Current scanned input value
- $h[n-k]$: System response shifted from k to n
- $n-k$: Input to output time gap
- $x[k]\,h[n-k]$: Contribution from input k

We call this "x convolved with h"

$$y[n] = x * h$$

* convolved

Algebraic Properties of Convolution

identity element

① $x * \delta = x$

② $x * h = h * x$ [Commutative]

③ $h_1 * (h_2 * h_3) = (h_1 * h_2) * h_3$ [Associative]

④ $h_1 * (h_2 + h_3) = (h_1 * h_2) + (h_1 * h_3)$ [Distributive]

...

System Properties extracted from h[n]

Since h[n] tells us the behaviour that every input sample will inherit. once h[n] is known, we can often determine important properties of the entire LTI system without testing many different inputs.

① Finite Impulse Response (FIR)

A system is FIR if its impulse response is non-zero for only a finite number of time indices.

Essentially, one impulse has a response of finite duration.

FIR : $h[n] \neq 0$ for only finitely many n

② Infinite Impulse Response (IIR)

A system is IIR if its impulse response remains non-zero for infinitely many time indices.

Essentially, one impulse has a response of infinite duration

FIR : $h[n] \neq 0$ for only infinitely many n

③ Memoryless

A memoryless system's output at time n depends only on the input at that same time n.

As learned previously, $y[n] = \sum_k x[k]\,h[n-k]$, normally many different k's could contribute to an output at a particular time.

But for a memoryless system, every contribution except $k = n$ must disappear.

Let $m = n - k$ — output time n, measured relative to time k when the impulse occured.

- $m < 0$: Output time is before input impulse occured
- $m = 0$: Output time is exactly when input impulse occured
- $m > 0$: Output time is after input impulse occured

$$h[m] = 0 \quad \text{for all} \quad m \neq 0$$
$$\downarrow$$
$$h[m] = K\,\delta[m] \quad \text{for some constant } K$$

Do note that the course notes reused the "n" notation to represent n-k.

④ Causal

A causal system may use the current or/and past inputs, but a future input cannot affect the output now.

Now look at convolution:

$$y[n] = \sum_k x[k]\,h[n-k]$$

A future input sample [Relative to $y[n]$] is $x[k]$ when $k > n$

$\Longrightarrow n - k < 0$

$\Longrightarrow m < 0$

$$h[m] = 0 \quad \text{for every} \quad m < 0$$

Do note that the course notes reused the "n" notation to represent n-k.

⑤ BIBO stability

Bounded Input, $x[k]$ produces only bounded output, $y[n]$

$$y[n] = \sum_k x[k]\, h[n-k]$$

Since $x[k]$ is assumed to be bounded, we only need to make sure that $\sum_k h[n-k]$ is also bounded.

$$\sum_{k=-\infty}^{\infty} |h[n-k]| = \sum_{k=-\infty}^{\infty} |h[k]|$$

$$\downarrow$$

$$\boxed{\sum_{k=-\infty}^{\infty} |h[k]| < \infty}$$

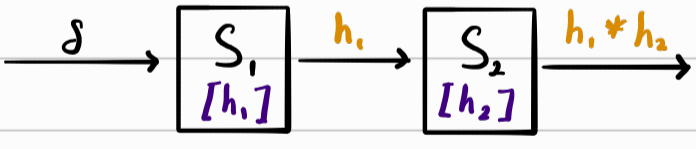
We call this "$h[n]$ is absolutely summable"

Essentially stability is just asking:

> Does one impulse response have a finite total amount of magnitude to distribute?

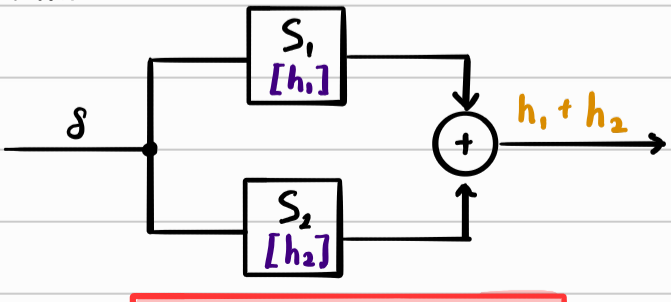
Interconnected LTI systems

① Series



$$h_{series} = h_1 * h_2$$

② Parallel



$$h_{parallel} = h_1 + h_2$$

Graphical DT convolution

We already understand that

$$y[n] = \sum_k x[k]\, h[n-k]$$

just means

> For one chosen output time n, scan through every input location k, find how much each input sample $x[k]$ contributes to the target output through $h[n-k]$, then add all the contributions

Now graphical convolution is just a visual method for carrying out convolution. The fllwing is the transformation that take place when finding one particular output $y[n]$ ← n is fixed

Reflect on $k=0$ → Shift right by n → Scale vertically by $x[k]$

$h[k] \quad h[-k] \quad h[n-k] \quad x[k]\,h[n-k]$ → Sum

$$y[n] = (x*h)[n] = \sum_k x[k]\, h[n-k]$$

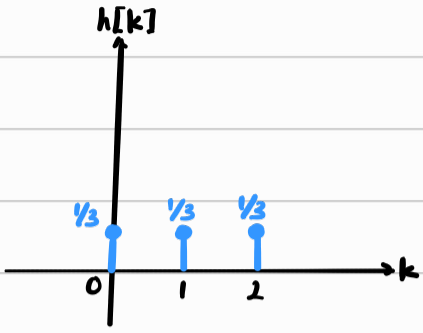
Essentially to get one particular output, $y[n]$ at a fixed n

Flip horizontally
↓
Shift right by n
↓
Multiply
↓
Sum

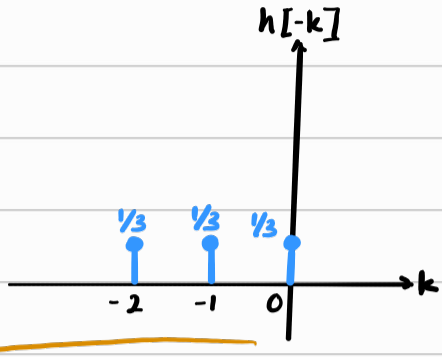
Example

Using the 3-point avg we have the following

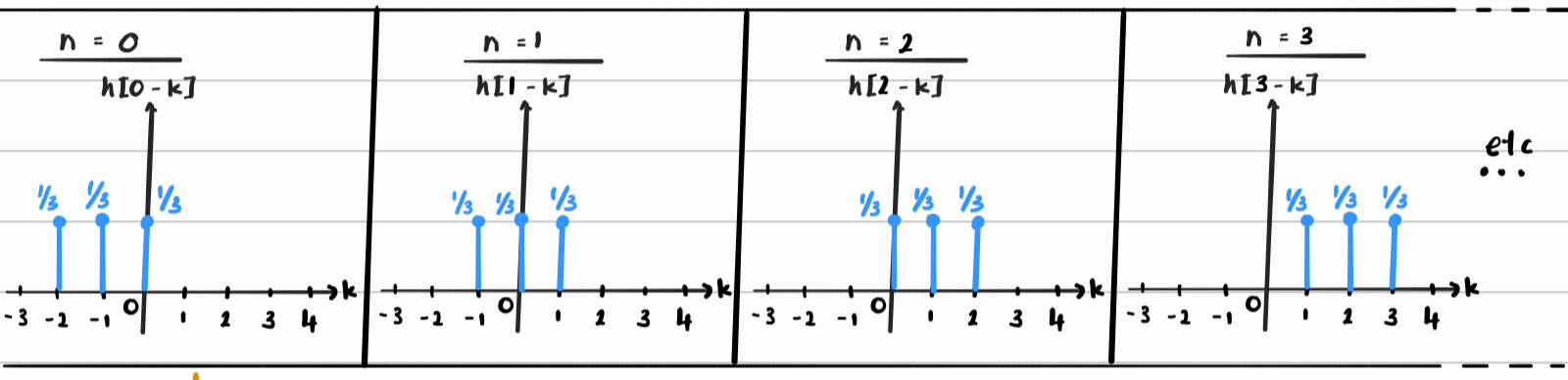
$x[k] = 3\delta[k] - \delta[k-1] + 2\delta[k-2]$; $h[k] = \frac{1}{3}(\delta[k] + \delta[k-1] + \delta[k-2])$



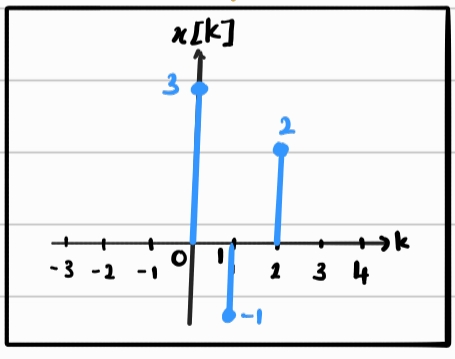
Flip about $k=0$
$h[-k]$

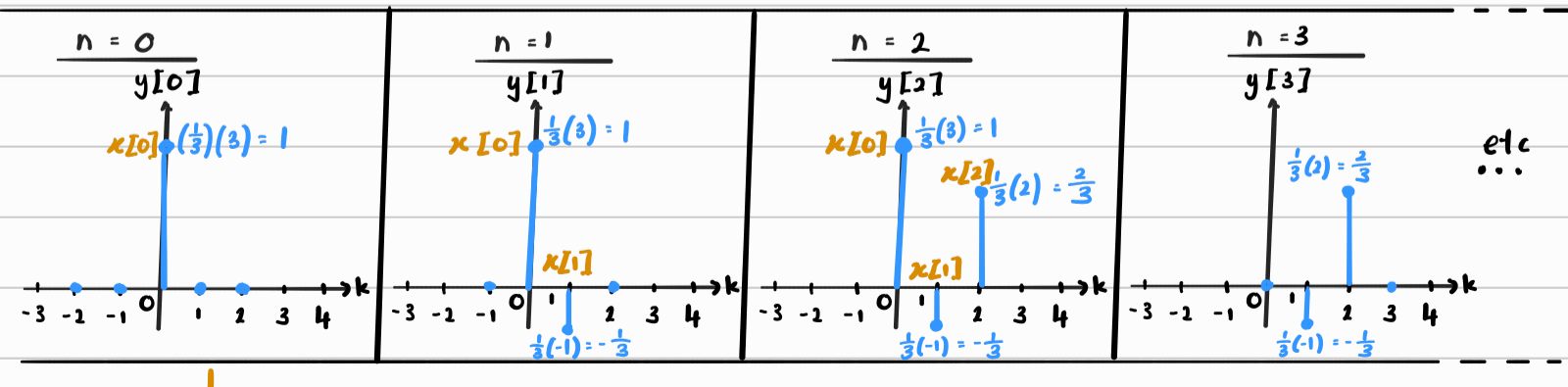


Shift right by n



Scale vertically by $x[k]$





Sum [Nxt Pg]

Table Form

	$x[0]$	$x[1]$	$x[2]$	
	$k=0$	$k=1$	$k=2$	Sum
$n=0$	1	0	0	$y[0] = 1$
$n=1$	1	$-1/3$	0	$y[1] = 1 + (-\frac{1}{3}) = \frac{2}{3}$
$n=2$	1	$-1/3$	$2/3$	$y[2] = 1 + (-\frac{1}{3}) + \frac{2}{3} = \frac{4}{3}$
$n=3$	0	$-1/3$	$2/3$	$y[3] = -\frac{1}{3} + \frac{2}{3} = \frac{1}{3}$
$n=4$	0	0	$2/3$	$y[4] = \frac{2}{3}$
$n=5$	0	0	0	$y[5] = 0$
$\vdots$		$\vdots$		$\vdots$

$$\therefore\ y[n] = \delta[n] + \frac{2}{3}\delta[n-1] + \frac{4}{3}\delta[n-2] + \frac{1}{3}\delta[n-3] + \frac{2}{3}\delta[n-4] \quad \#$$

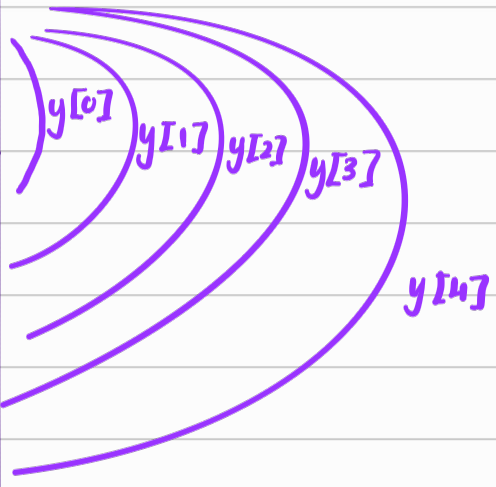
Sequence Table Method

Still using the same 3-point avg where

$x[n] = 3\delta[n] - \delta[n-1] + 2\delta[n-2]$

$h[n] = \frac{1}{3}(\delta[n] + \delta[n-1] + \delta[n-2])$

	k	-2	-1	0	1	2	3	4
	x[k]	0	0	3	-1	2	0	0
	h[k]	0	0	1/3	1/3	1/3	0	0
n=0	h[-k]	1/3	1/3	1/3	0	0	0	0
n=1	h[1-k]	0	1/3	1/3	1/3	0	0	0
n=2	h[2-k]	0	0	1/3	1/3	1/3	0	0
n=3	h[3-k]	0	0	0	1/3	1/3	1/3	0
n=4	h[4-k]	0	0	0	0	1/3	1/3	1/3



$y[0] = x[k]\,h[-k]$

$= 3(\frac{1}{3})$

$y[0] = 1$ #

$y[1] = x[k]\,h[1-k]$

$= 3(\frac{1}{3}) + -1(\frac{1}{3})$

$= 2/3$ #

$y[2] = x[k]\,h[2-k]$

$= 3(\frac{1}{3}) + -1(\frac{1}{3}) + 2(\frac{1}{3})$

$= 4/3$ #

$y[3] = x[k]\,h[3-k]$

$= -1(\frac{1}{3}) + 2(\frac{1}{3})$

$= \frac{1}{3}$ #

$y[4] = x[k]\,h[4-k]$

$= 2(\frac{1}{3})$

$= \frac{2}{3}$ #

Continuous-Time Convolution

The core mechanism is exactly the same, just that now we are dealing with continuous time, so we have to scan not only integer sample locations, but rather continuum of real-valued time location.

$$x[n] = \sum_k x[k]\,\delta[n-k] \xrightarrow{CT} x(t) = \int_{-\infty}^{\infty} x(\tau)\,\delta(t-\tau)\,d\tau$$

$$y[n] = \sum_k x[k]\,h[n-k] \xrightarrow{CT} y(t) = \int_{-\infty}^{\infty} x(\tau)\,h(t-\tau)\,d\tau$$

An important thing to note is that now,

$y(n)$ = signed area under $x(\tau)\,h(t-\tau)$

it isn't a singular discrete data point anymore.

The biggest difference is that in DT, shifted impulse response $h[n-k]$ moves only btwn integer locations, so the overlap changes in discrete steps where we can often calc. each output sample directly.

In CT, t can take infinately many real values, so the overlap changes continuously. So we have to integrate & find the boundaries of integration.

Reminder:

t : target output time.

When finding one particular output value, $y(t)$, treat t as fixed

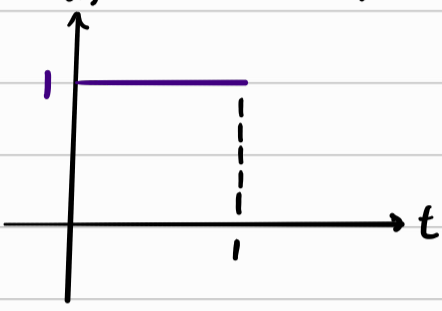
τ : Continuous input-time scanner

It runs through all possible input times & accumulate their contribution to the chosen output time t.

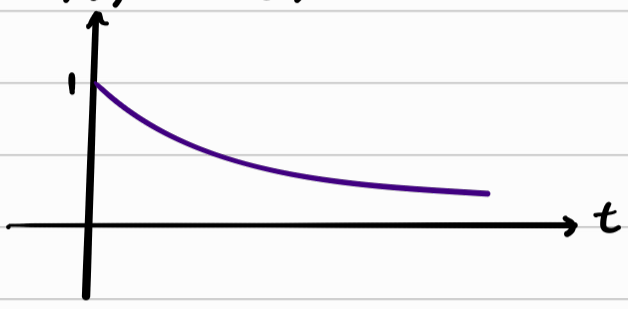
Example

Find $y(t) = x * h$ where ...

$h(t) = u(t) - u(t-1)$



$x(t) = e^{-t}u(t)$



Answer

$x(\tau) = e^{-\tau}, \quad \tau > 0 \quad$ —— ①

Since $h(t-\tau) \neq 0$ when $0 \leq t-\tau \leq 1$, solving for τ we get ...

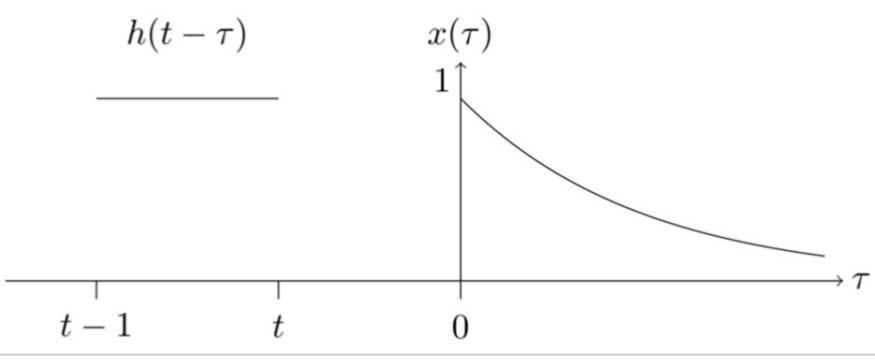
$$h(t) = \begin{cases} 1, & 0 \leq t \leq 1 \\ 0, & \text{otherwise} \end{cases} \Longrightarrow h(t-\tau) \neq 0 \text{ for } t-1 \leq \tau \leq t$$

Case 1 : No overlap $\Rightarrow t \leq 0$

τ is scanning through $t-1 \leq \tau \leq t$

Since $x(\tau)$ only exist for $\tau \geq 0$

$\Longrightarrow y(t) = 0, \quad t \leq 0$



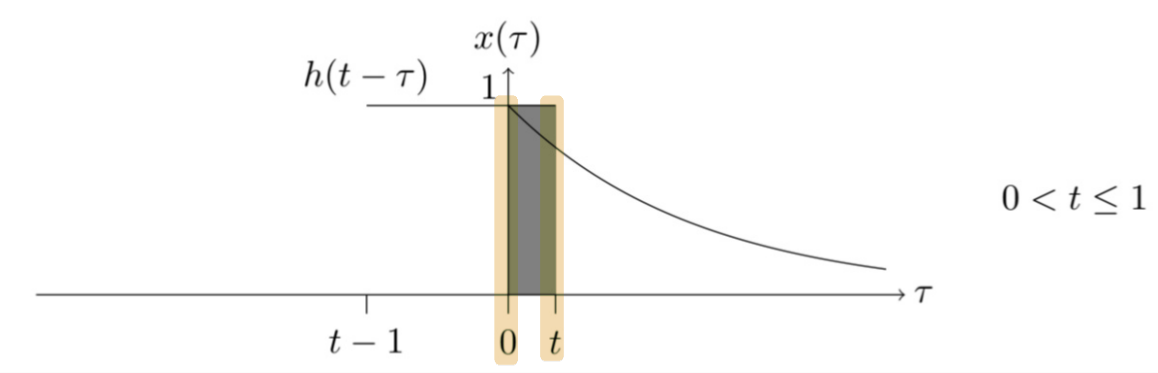
Case 2 : Partial Overlap $\Rightarrow 0 < t \leq 1$

τ is scanning through $t-1 \leq \tau \leq t$

Since $x(\tau)$ only exist for $\tau \geq 0$

$\Rightarrow$ Partial overlap is $0 \leq \tau \leq t$

$\Rightarrow y(t) = \int_0^t e^{-\tau} d\tau = 1 - e^{-t}, \quad 0 < t \leq 1$ #

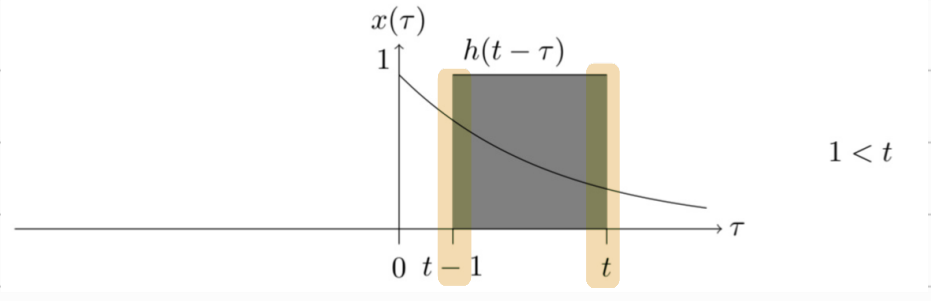


Case 3 : Complete overlap $\Rightarrow t > 1$

τ is scanning through $t-1 \leq \tau \leq t$

Since $x(\tau)$ exist for $\tau \geq 0$

$\Rightarrow y(t) = \int_{t-1}^t e^{-\tau} d\tau = (e-1)e^{-t}, \quad t > 1$ #



$\therefore y(t) = \begin{cases} 0, & t \leq 0 \\ 1 - e^{-t}, & 0 < t \leq 1 \\ e^{1-t} - e^{-t}, & t > 1 \end{cases}$ #

Reminder :

The value t determines where the shifted impulse response $h(t-\tau)$ lies. The actual τ range we integrate over is where this window overlaps with the non-zero region of $x(\tau)$

Alternative way to describe CT LTI system

So far we have been probing a system using

unit impulse, $\delta(t)$

and observing

impulse response, $h(t)$

We can actually also probe the system using

unit step, $u(t)$

and observing

step response, $s(t)$

$$u(t) \xrightarrow{S} s(t)$$

Connection btwn $s(t)$ and $h(t)$

Recall that...

$$u(t) = \int_{-\infty}^{t} \delta(\tau)\, d\tau$$

↓ System is linear

$$s(t) = \int_{-\infty}^{t} h(\tau)\, d\tau$$

OR

$$h(t) = \frac{d}{dt} s(t)$$

DT vs CT LTI Summary Table

Conceptual job	Discrete Time (DT)	Continuous Time (CT)
Unit impulse	$\delta[n]$	$\delta(t)$
Key impulse idea	Height 1 at one sample	Unit **area** concentrated at one time
Sifting formula	$x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k]$	$x(t) = \int_{-\infty}^{\infty} x(\tau)\delta(t-\tau)\,d\tau$
Dummy variable	k	τ
Impulse response	$\delta[n] \to h[n]$	$\delta(t) \to h(t)$
LTI response to shifted impulse	$\delta[n-k] \to h[n-k]$	$\delta(t-\tau) \to h(t-\tau)$
Scaled + shifted impulse response	$x[k]\delta[n-k] \to x[k]h[n-k]$	$x(\tau)\delta(t-\tau) \to x(\tau)h(t-\tau)$
Output / convolution	$y[n] = (x*h)[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$	$y(t) = (x*h)(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)\,d\tau$
Convolution calculation	Flip → shift → multiply → **sum**	Flip → shift → multiply → **integrate**
Identity	$x * \delta = x$	$x * \delta = x$
Commutative	$x * h = h * x$	$x * h = h * x$
Series / cascade	$h_{\text{series}} = h_1 * h_2$	$h_{\text{series}} = h_1 * h_2$
Parallel	$h_{\text{parallel}} = h_1 + h_2$	$h_{\text{parallel}} = h_1 + h_2$
Memoryless	$h[n] = \alpha\delta[n]$	$h(t) = \alpha\delta(t)$
Causal	$h[n] = 0,\ n < 0$	$h(t) = 0,\ t < 0$
BIBO stable	$\sum_{k=-\infty}^{\infty} \lvert h[k]\rvert < \infty$	$\int_{-\infty}^{\infty} \lvert h(\tau)\rvert\,d\tau < \infty$
Name of stability condition	$h[n]$ is **absolutely summable**	$h(t)$ is **absolutely integrable**

WEEK 4

Fourier Series

Time Domain to Frequency Domain

Previously we decompose and understand complicated signals using the time-domain building block, $\delta[n]$

Now we do the same thing but by using the frequency-domain building block, $e^{j(\omega t)}$

Recall from week 1 : Unit Complex Exponential $x(t) = e^{j(\omega t)}$

Using Euler's Formula,

$$e^{j(\omega t)} = \cos(\omega t) + j\sin(\omega t)$$

and as a result, ordinary sinusoids can be represented using complex exponentials.

$$\sin(\omega t) = \frac{1}{2j}\left[e^{j(\omega t)} - e^{-j(\omega t)}\right]$$

$$\cos(\omega t) = \frac{1}{2}\left[e^{j(\omega t)} + e^{-j(\omega t)}\right]$$

The useful idea here is that complicated periodic signals (complex waves) is actually multiple sinusoids added together.

∴ We can essentially decompose / describe a complicated signal into a sum of sinusoids.

Periodicity of Sums of Sinusoids

We want to build periodic signals by adding complex exponentials.

One important condition is :

> For the whole sum to repeat, all of its components must eventually repeat tgt

Continuous-Time Case

Consider $x(t) = Ae^{j(\omega_1 t)} + Be^{j(\omega_2 t)}$ with individual fundamental periods of $T_1 = \frac{2\pi}{|\omega_1|}$; $T_2 = \frac{2\pi}{|\omega_2|}$ —— ①

Each term has many valid periods :

$$T_1, 2T_1, 3T_1, \ldots$$

and

$$T_2, 2T_2, 3T_2, \ldots$$

The sum $x(t)$ repeats only if we can find one time T that is a period of both terms.

∴ We require positive k_1, k_2 such that

$$T = k_1 T_1 = k_2 T_2$$

The smallest such common T is the fundamental period, T_0 of the whole signal

Using Eqn ①,

$$k_1 \left[\frac{2\pi}{\omega_1}\right] = k_2 \left[\frac{2\pi}{\omega_2}\right]$$

$$\frac{\omega_1}{\omega_2} = \frac{k_1}{k_2}$$

$$\frac{\omega_1}{\omega_2} \in Q$$

Harmonic frequency is any frequency that is an integer multiple of the fundamental frequency. $[f_n = n \times f_0]$

(n: harmonic number)

∴ Sum of sinusoids is periodic when frequencies are harmonically related

This then follows that...

$$\text{Fundamental Angular Frequency}, \omega_0 = \frac{2\pi}{T_0}$$

Discrete-Time Case

Recall that a DT complex exponential, $e^{j(\omega n)}$ is NOT automatically periodic.

We need a positive integer N such that

$$e^{j\omega(n+N)} = e^{j\omega n}$$

$$\downarrow$$

$$e^{j(\omega N)} = 1$$

which then requires

$$\omega N = (2\pi)m \quad m \in \mathbb{Z} \longrightarrow \frac{2\pi}{\omega} = \frac{N}{m} \in Q$$

∴ For an individual DT sinusoid to be periodic, $\frac{2\pi}{\omega}$ must be rational

Once we have two periodic signals with period N_1, N_2, their sum is periodic with a common period

$$N = LCM(N_1, N_2)$$

Short Summary

An individual CT is always periodic, but a sum of CT sinusoids is only periodic if the ratio of their frequency is rational $\left(\frac{\omega_1}{\omega_2} \in Q\right)$.

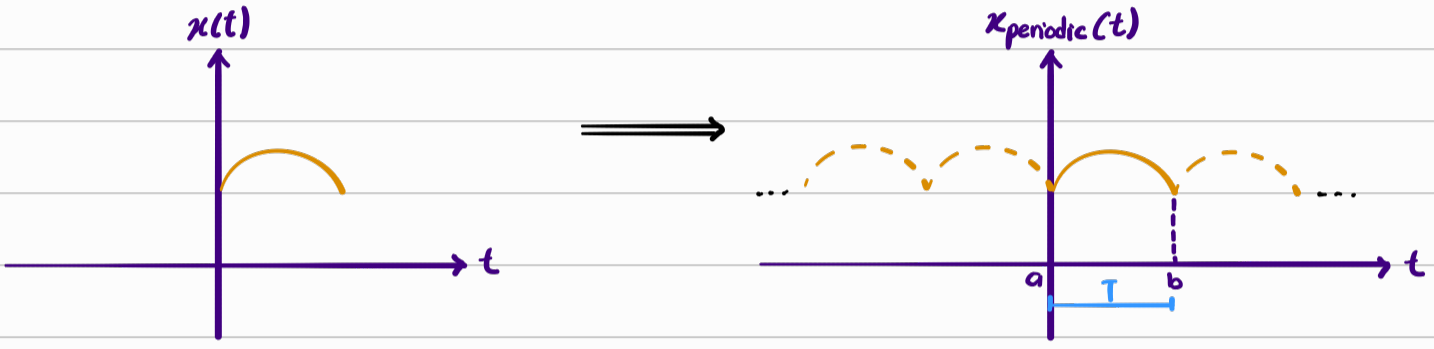
Conversely...

An individual DT is only periodic if $\left(\frac{\omega}{2\pi} \in Q\right)$, but sums of periodic sinusoids is always periodic

Periodic Extension of Finite Duration Signals

Generally, periodic signals goes on indefinately, however we also want to decompose/describe practical signals that only exist for some finite duration. Thus, we just treat the finite duration signal as the period itself that keeps repeating (but only look at our original signal interval)

① Continuous-Time Periodic Extension



$$x_{periodic}(t) = \sum_{m=-\infty}^{\infty} x(t - mT)$$

shifted copies by integer multiple of T

, where $T = b - a$

② Discrete-Time Periodic Extension

$$x_{periodic}[n] = \sum_{m=-\infty}^{\infty} x[n - mN]$$

shifted copies by integer multiple of N

, $N = B - A + 1$

Continuous-Time Fourier Series

Fourier Series says that a sufficiently "well-behaved" periodic signal can be constructed by adding scaled copies of these harmonics.

Fourier Synthesis

$$x(t) = \sum_{k=-\infty}^{\infty} X_k e^{j(k\omega_0 t)}$$

- $e^{j(k\omega_0 t)}$: k-th frequency building block
- X_k: Amount of that building block contained in the signal

So one term = one frequency's contribution. Summing up all the harmonic frequencies contribution creates/sythesises the complete signal, thus given the name Fourier Synthesis

Fourier Coefficient, X_k

X_k is generally a complex number. This is so that the building block, $e^{j(k\omega_0 t)}$ can be "phase shifted" as needed

$$X_k = |X_k| e^{j\angle X_k}$$

- $\angle X_k$: Phase of harmonic k
- $|X_k|$: Magnitude of harmonic k

Since the building block is already known as $e^{j(\omega_0 t)}, e^{j(2\omega_0 t)}, e^{j(3\omega_0 t)} \ldots$ knowing the set of all the coefficients is all we need to reconstruct $x(t)$ into the frequency domain. Thus we just call the frequency domain with the following notation...

$$x(t) \longrightarrow \{X_k\}$$

- $x(t)$: Time domain
- $\{X_k\}$: Frequency domain

Fourier Analysis

Fourier Synthesis answers $\{X_k\} \rightarrow x(t)$ [Given the components, build the signal]

Fourier Analysis answers the reverse $x(t) \rightarrow \{X_k\}$

[Given the signal, find the components]

$$X_k = \frac{1}{T_0} \int_a^{a+T_0} x(t)\, e^{-j(k\omega_0 t)}\, dt$$

Accumulates comparison over one period

compares the signal against the selected k-th frequency

Essentially, X_k = measure of how strongly harmonic k exists in $x(t)$.

The "comparator term" works via orthogonality.

Lets say we want to extract a frequency of "k" out of the signal

$$x(t)\, e^{-j(k\omega_0 t)} = [X_l\, e^{j(l\omega_0 t)}]\, e^{-j(k\omega_0 t)}$$

$$= X_l\, e^{j(l-k)\omega_0 t}$$

where l = harmonic currently being scanned in the sum

$$\frac{1}{T_0} \int_a^{a+T_0} e^{j(l-k)\omega_0 t}\, dt = \begin{cases} 1, & l = k \\ 0, & l \neq k \end{cases}$$

Frequency matched

Frequency not matched

This causes all unwanted harmonics to vanish, leaving only X_k.

DC Component

Vertical offset / Avg value

This term is the first term of the Fourier Series, [k = 0]

$$e^{-j(0\omega_0 t)} = 1$$

$$\downarrow$$

$$X_0 = \frac{1}{T_0} \int_a^{a+T_0} x(t)\, dt$$

This is simply the constant average value of the signal over one period.

Finding Fourier Coefficients by Inspection

The Fourier Analysis Integral formula is the general method. However, if the signal is already written as a sum of harmonically related sinusoids / complex exponential, its just easier to use Euler's and read off X_k.

Using $\cos\theta = \frac{1}{2}(e^{j\theta} + e^{-j\theta})$
$\sin\theta = \frac{1}{2j}(e^{j\theta} - e^{-j\theta})$

For example,

$$x(t) = 2\cos(2t) + 3\sin(t) - 1$$
$$= e^{j(-2t)} + \frac{3j}{2}e^{j(-t)} - 1e^{j(0)} - \frac{3j}{2}e^{j(t)} + e^{j(2t)}$$

Comparing to $x(t) = \sum_k X_k e^{j(k\omega_0 t)}$ where $\omega_0 = 1$

$\Rightarrow X_{-2} = 1,\ X_{-1} = \frac{3j}{2},\ X_0 = -1,\ X_1 = -\frac{3j}{2},\ X_2 = 1$ #

CT Fourier Series Properties

Once we know X_k, we often don't need to redo the analysis integral every time $x(t)$ changes. Many time-domain operations corresponds to simple changes in the Fourier Coefficient.

① Linearity

If $y(t) = \alpha x(t) + \beta z(t)$ with corresponding coefficients: X_k, Z_k, Y_k

$$Y_k = \alpha X_k + \beta Z_k$$

② Modulation

Suppose $y(t) = e^{j(M\omega_0 t)}x(t)$ where M is an integer

Then,

$$y(t) = \sum_k X_k e^{j(k+M)\omega_0 t}$$

$$Y_k = X_{k-M}$$

③ Multiplication

If $y(t) = [x(t)][z(t)]$

One contribution to Y_k

$$Y_k = \sum_{l=-\infty}^{\infty} X_l Z_{k-l}$$

This is convolution of the Fourier Coefficients

where k : target frequency

l : Frequency Coefficient Scanner

Multiplication in time ⟷ Convolution in frequency

④ Periodic Convolution

If $y(t) = \int_0^{T_0} x(\tau)\, z(t-\tau)\, d\tau = x * z$

$$Y_k = T_0 [X_k Z_k]$$

Periodic convolution in time ⟷ Multiplication in frequency

⑤ Conjugation

If $y(t) = x^*(t)$

$$Y_k = X^*_{-k}$$

If $x(t)$ is purely real ↓ $x^*(t) = x(t)$

$$X_k = X^*_{-k}$$

⑥ Differentiation

If $y(t) = \frac{d}{dt} x(t)$,

Then $y(t) = \frac{d}{dx}\left[\sum_k X_k e^{j(k\omega_0 t)}\right]$

$= \sum_k (jk\omega_0) X_k e^{j(k\omega_0 t)}$

$$Y_k = (jk\omega_0) X_k$$

Parseval's Relation

This just gives a representation of power, P_x in the frequency-domain

$$P_x = \frac{1}{T_0}\int_{T_0}^{a+T_0} |x(t)|^2 \, dt = \sum_{k=-\infty}^{\infty} |X_k|^2$$

Time-Domain | Frequency Domain

Essentially its saying that one frequency component $[X_k e^{j(k\omega_0 t)}]$ has avg power $|X_k|^2$

Fourier Approximation

A CT Fourier Series may contain infinately many harmonic components.

$$x(t) = \sum_k X_k e^{j(k\omega_0 t)}$$

But often the largest coefficients contain most of the important information.

So instead of keeping every harmonic, keep only

$$-N \leq k \leq N$$

$$x_N(t) = \sum_{k=-N}^{N} X_k e^{j(k\omega_0 t)}$$

And as expected, as $N \to \infty$, $x_N(t) \longrightarrow x(t)$

CT Fourier Series Convergence small detail

- If $x(t)$ is continuous at the point [Fourier Series $\rightarrow x(t)$]
- If $x(t)$ has a finite jump discontinuity
  [Fourier Series $\rightarrow \frac{x(t^-) + x(t^+)}{2}$]

that is the midpoint of the left and right-hand values.

Approximation Error

Define

$$e_N(t) = x(t) - x_N(t)$$

Since x_N covers $-N$ to N, the error only contains...

$$k < -N \text{ or } k > N \implies |k| > N$$

Using Parseval we get

$$P_{e_N} = P_x - \sum_{k=-N}^{N} |X_k|^2$$

Discrete-Time Fourier Series DTFS

The mechanism of DTFS is conceptually identical to CT.
The crucial difference is that unlike CT (which has an infinite number of unique harmonics), DT only has N_0 unique harmonics.

Because time n is strictly an integer, increasing the harmonic index k by exactly N_0 shifts the frequency by 2π, this causes the samples to land on the exact same values.

e.g. If $N_0 = 4$, harmonics 0, 4, 8, 12 ...
harmonics 1, 5, 9, 13 ...
⋮
are all the exact same wave

So DT Fourier Series summation limits its search to just one complete cycle of unique harmonics.

DT Fourier Synthesis

$$x[n] = \sum_{k=0}^{N_0-1} X_k e^{j(k\omega_0 n)}$$

DT Fourier Analysis

$$X_k = \frac{1}{N_0} \sum_{n=0}^{N_0-1} x[n] e^{-j(k\omega_0 n)}$$

where k = fixed target harmonic
n = sample scanner through one period.

Properties of DTFS are the same as CTFS

ECE 2111 Week 4 — Fourier Series Terminology & Formula Table

Terminology / notation	Formula	Concise explanation
CT unit complex exponential	$x(t) = e^{j\omega_0 t}$	Basic CT frequency-domain building block; ω_0 is angular frequency in rad/s.
Scaled / phase-shifted CT complex exponential	$x(t) = re^{j\theta}e^{j\omega_0 t}$	r sets magnitude and θ sets phase.
Real part of a scaled CT complex exponential	$x(t) = r\cos(\omega_0 t + \theta)$	The corresponding real sinusoid.
Euler's identity	$e^{j\theta} = \cos\theta + j\sin\theta$	Connects complex exponentials with ordinary sinusoids.
Cosine in complex-exponential form	$\cos\theta = \frac{1}{2}(e^{j\theta} + e^{-j\theta}) = \mathrm{Re}[e^{j\theta}]$	Represents a cosine using positive- and negative-frequency exponentials.
Sine in complex-exponential form	$\sin\theta = \frac{1}{2j}(e^{j\theta} - e^{-j\theta}) = \mathrm{Im}[e^{j\theta}]$	Represents a sine using positive- and negative-frequency exponentials.
CT frequency / period units	$\lvert\omega_0\rvert = 2\pi\lvert f_0\rvert = \frac{2\pi}{T_0}$	f_0 is in cycles/s (Hz), ω_0 in rad/s, and T_0 is positive.
CT other valid periods	$T = \frac{2\pi k}{\omega_0}, \quad k = 1, 2, 3, \ldots$	For positive fundamental angular frequency, any positive integer multiple of T_0 is also a period.
Two-frequency CT complex-exponential signal	$x(t) = Ae^{j\omega_1 t} + Be^{j\omega_2 t}$	General two-component form used to study when a sum is periodic.
Common-period condition for two CT components	$T = k_1 T_1 = k_2 T_2$	A common period completes an integer number of cycles of both components.
Common period written using frequencies	$T = \frac{2\pi k_1}{\omega_1} = \frac{2\pi k_2}{\omega_2}$	Equivalent common-period condition written directly in terms of the two angular frequencies.
Harmonically related CT frequencies	$\frac{\omega_1}{\omega_2} = \frac{k_1}{k_2} \in \mathbb{Q}$	The frequency ratio must be rational for the two components to share a common period.
Orthogonality of CT harmonics	$\frac{1}{T_0}\int_0^{T_0} e^{j\ell\omega_0 t}e^{-jk\omega_0 t}\,dt = 1\ (\ell = k),\ 0\ (\ell \neq k)$	Matching harmonics survive the one-period average; different harmonics cancel.
DT unit complex exponential	$x[n] = e^{j\omega n}$	Basic DT frequency-domain building block; ω is in rad/sample.
Scaled / phase-shifted DT complex exponential	$x[n] = re^{j\theta}e^{j\omega n}$	DT complex exponential with magnitude r and phase θ.
Real part of a scaled DT complex exponential	$x[n] = r\cos(\omega n + \theta)$	The corresponding real DT sinusoid.
DT complex-exponential periodicity condition	$N = m\frac{2\pi}{\omega}, \quad m, N \in \mathbb{Z}_{>0}$	A DT complex exponential is periodic only when a positive integer sample period N exists.
DT fundamental angular frequency	$\omega_0 = \frac{2\pi}{N_0}$	N_0 is the smallest positive period in samples.

Terminology / notation	Formula	Concise explanation
DT frequency / period units	$\omega_0 = 2\pi f_0 = \frac{2\pi}{N_0}$	f_0 is in cycles/sample; the lecture-note units box writes T_0 in its final fraction, while the DT period relation elsewhere uses N_0.
DT other valid periods	$N = kN_0, \quad k = 1, 2, 3, \ldots$	Positive integer multiples of the fundamental DT period are also periods.
Minimum DT period / maximum fundamental frequency	$N_{0,\,\min} = 2 \Rightarrow \omega_{0,\,\max} = \pi$ rad/sample	Non-constant DT periodic signals cannot have a fundamental period below two samples.
Maximum-frequency DT prototype	$x[n] = (-1)^n = e^{j\pi n}$	Alternates between $+1$ and -1; it has period 2 and frequency π rad/sample.
Sum of two periodic DT signals	$N_0 = \mathrm{LCM}\,(N_1, N_2)$	For component fundamental periods N_1, N_2, the sum repeats with their least common multiple.
CT finite-duration support	$x(t) = 0 \quad \text{for } t < a \text{ and } t > b$	The CT signal is nonzero only on the finite interval $[a, b]$.
CT finite duration	$T = b - a$	Duration of a CT signal supported from a to b.
DT finite-duration support	$x[n] = 0 \quad \text{for } n < A \text{ and } n > B$	The DT signal is nonzero only for samples $A, \ldots, B$.
DT finite duration	$N = B - A + 1$	Number of samples from A through B, including both endpoints.
CT periodic extension	$x_{\text{periodic}}(t) = \sum_{m=-\infty}^{\infty} x(t - mT)$	Repeats a finite CT signal every $T = b - a$ seconds by shift-and-add.
DT periodic extension	$x_{\text{periodic}}[n] = \sum_{m=-\infty}^{\infty} x[n - mN]$	Repeats a finite DT signal every $N = B - A + 1$ samples by shift-and-add.
CT harmonic building block	$e^{jk\omega_0 t}, \quad k \in \mathbb{Z}$	k is the harmonic/frequency index; the actual angular frequency is $k\omega_0$.
CT Fourier-series synthesis	$x(t) = \sum_{k=-\infty}^{\infty} X_k e^{jk\omega_0 t}$	Reconstructs a periodic CT signal from all of its Fourier coefficients.
CT Fourier-series analysis	$X_k = \frac{1}{T_0}\int_0^{T_0} x(t)\, e^{-jk\omega_0 t}\, dt$	Finds the coefficient associated with harmonic k by integrating over one period.
CT Fourier analysis over any period	$X_k = \frac{1}{T_0}\int_a^{a+T_0} x(t)\, e^{-jk\omega_0 t}\, dt$	The integration window may start anywhere as long as it spans exactly one full period.
CT DC / average coefficient	$X_0 = \frac{1}{T_0}\int_0^{T_0} x(t)\, dt$	X_0 is the average value of the signal over one period.
Uniqueness of CT Fourier coefficients	$\sum_{k=-\infty}^{\infty} X_k e^{jk\omega_0 t} = \sum_{k=-\infty}^{\infty} Z_k e^{jk\omega_0 t} \Rightarrow X_k = Z_k$	A periodic signal has a unique set of Fourier coefficients.
Fourier coefficient in polar form	$X_k = \lvert X_k \rvert e^{j\angle X_k}$	Each coefficient contains a magnitude and a phase.

Terminology / notation	Formula	Concise explanation
Coefficient magnitude	X_k	Magnitude/strength of the k-th harmonic in the frequency-domain representation.
Coefficient phase	$\angle X_k$	Phase associated with the k-th harmonic.
Truncated CT Fourier series	$x_N(t) = \sum_{k=-N}^{N} X_k e^{jk\omega_0 t}$	Finite partial sum used to approximate a periodic CT signal.
CT Fourier-series convergence interpretation	$\lim_{N\to\infty} \sum_{k=-N}^{N} X_k e^{jk\omega_0 t}$	The infinite synthesis sum is interpreted as the limit of its partial sums where convergence holds.
Dirichlet absolute-integrability condition	$\int_0^{T_0} \lvert x(t) \rvert dt \le M < \infty$	One Dirichlet condition ensuring the Fourier coefficients are finite and convergence is well behaved.
Finite-duration CT Fourier frequency	$\omega_0 = \frac{2\pi}{b-a}$	Frequency spacing used for the periodic extension of a finite CT signal on $[a, b]$.
Finite-duration CT synthesis	$x(t) = \sum_{k=-\infty}^{\infty} X_k e^{jk\omega_0 t}, \quad a \le t \le b$	On the original interval, the periodic extension's Fourier series equals the finite-duration signal.
Finite-duration CT Fourier coefficients	$X_k = \frac{1}{b-a}\int_a^b x(t)\, e^{-jk\omega_0 t}\, dt$	Computes the coefficients directly from the finite CT interval.
Trigonometric CT Fourier-series synthesis	$x(t) = a_0 + \sum_{k=1}^{\infty} \left[a_k \cos\left(k\omega_0 t\right) + b_k \sin\left(k\omega_0 t\right)\right]$	Optional sine/cosine form of the same CT Fourier-series representation for real signals.
Trigonometric DC coefficient	$a_0 = \frac{1}{T_0}\int_0^{T_0} x(t)\, dt$	Constant term in the trigonometric Fourier series.
Trigonometric cosine coefficient	$a_k = \frac{2}{T_0}\int_0^{T_0} x(t)\, \cos\left(k\omega_0 t\right) dt$	Measures the cosine contribution at harmonic k, for $k = 1, 2, \ldots$.
Trigonometric sine coefficient	$b_k = \frac{2}{T_0}\int_0^{T_0} x(t)\, \sin\left(k\omega_0 t\right) dt$	Measures the sine contribution at harmonic k, for $k = 1, 2, \ldots$.
Complex / trigonometric DC relation	$a_0 = X_0$	The constant term is the same in both Fourier-series forms.
Complex / trigonometric cosine relation	$a_k = X_k + X_{-k}$	Relates the cosine coefficient to the positive- and negative-index complex coefficients.
Complex / trigonometric sine relation	$b_k = j\left(X_k - X_{-k}\right)$	Relates the sine coefficient to the positive- and negative-index complex coefficients.
Complex coefficient from trig coefficients	$X_k = \frac{a_k - jb_k}{2}, \quad k \ge 1$	For positive harmonics; $X_0 = a_0$ is handled separately.
Negative-index complex coefficient	$X_{-k} = \frac{a_k + jb_k}{2}, \quad k \ge 1$	Companion negative-frequency coefficient for a real trigonometric Fourier series.

Terminology / notation	Formula	Concise explanation
Fourier-series linearity	$y = \alpha x + \beta z \Rightarrow Y_k = \alpha X_k + \beta Z_k$	For CT or DT periodic signals with the same fundamental period, coefficients combine linearly.
CT modulation / frequency shifting	$y(t) = e^{jM\omega_0 t} x(t) \Rightarrow Y_k = X_{k-M}$	Multiplication by harmonic M shifts the Fourier-coefficient index by M.
CT periodic convolution	$y(t) = \int_0^{T_0} x(\tau) z(t-\tau)\, d\tau \Rightarrow Y_k = T_0 X_k Z_k$	Periodic convolution in time becomes coefficient-wise multiplication, scaled by T_0.
CT multiplication	$y(t) = x(t) z(t) \Rightarrow Y_k = \sum_{\ell=-\infty}^{\infty} X_\ell Z_{k-\ell}$	Multiplication in time becomes convolution of the CT Fourier-coefficient sequences.
CT conjugation	$y(t) = x^*(t) \Rightarrow Y_k = X^*_{-k}$	Conjugating the signal conjugates the coefficient and reverses its harmonic index.
Conjugate symmetry for real signals	$X_k = X^*_{-k}$	For real CT or DT signals, positive- and negative-index coefficients occur as conjugate pairs.
Magnitude symmetry for real signals	$X_k=X_{-k}$	The Fourier-series magnitude is even in the harmonic index for a real signal.
Phase symmetry for real signals	$\angle X_k = -\angle X_{-k}$	The Fourier-series phase is odd in the harmonic index for a real signal.
Average power of one CT harmonic	$\frac{1}{T_0}\int_0^{T_0} \lvert X_k e^{jk\omega_0 t}\rvert^2\, dt = \lvert X_k\rvert^2$	The magnitude-squared coefficient is the average power of that harmonic component.
CT average power per period	$P_x = \frac{1}{T_0}\int_0^{T_0} \lvert x(t)\rvert^2\, dt$	Average squared magnitude of a periodic CT signal over one period.
CT Parseval relation	$\frac{1}{T_0}\int_0^{T_0} \lvert x(t)\rvert^2\, dt = \sum_{k=-\infty}^{\infty} \lvert X_k\rvert^2$	Total average power equals the sum of the powers of all Fourier components.
Magnitude-squared signal	$y(t) = x(t) x^*(t) = \lvert x(t)\rvert^2$	Special product used in the derivation of Parseval's relation.
Fourier coefficients of the magnitude-squared signal	$Y_k = \sum_{\ell=-\infty}^{\infty} X_\ell X^*_{\ell-k}$	For $y(t) = \lvert x(t)\rvert^2$, follows by combining multiplication with conjugation.
Average power as a DC coefficient	$\frac{1}{T_0}\int_0^{T_0} \lvert x(t)\rvert^2\, dt = Y_0$	If $y(t) = \lvert x(t)\rvert^2$, its zeroth Fourier coefficient equals the average power of x.
CT differentiation property	$y(t) = \frac{dx(t)}{dt} \Rightarrow Y_k = jk\omega_0 X_k = jk\frac{2\pi}{T_0} X_k$	Differentiation keeps the harmonic locations but scales each coefficient by $jk\omega_0$.
AM carrier multiplication	$y(t) = \cos(\omega_c t) x(t)$	General amplitude-modulation relationship shown as an application of frequency shifting.

Terminology / notation	Formula	Concise explanation
DT Fourier-series synthesis	$x[n] = \sum_{k=0}^{N_0-1} X_k e^{jk\omega_0 n}$	Reconstructs a period-N_0 DT signal from its N_0 distinct harmonics.
DT Fourier-series synthesis over any coefficient period	$x[n] = \sum_{k=a}^{a+N_0-1} X_k e^{jk\omega_0 n}$	Any N_0 consecutive coefficient indices may be used because X_k is periodic in k.
DT Fourier-series analysis	$X_k = \frac{1}{N_0}\sum_{n=0}^{N_0-1} x[n] e^{-jk\omega_0 n}$	Finds the k-th DT Fourier coefficient by summing over one full period.
DT Fourier analysis over any period	$X_k = \frac{1}{N_0}\sum_{n=a}^{a+N_0-1} x[n] e^{-jk\omega_0 n}$	The analysis window may be any N_0 consecutive samples.
DT DC / average coefficient	$X_0 = \frac{1}{N_0}\sum_{n=0}^{N_0-1} x[n]$	X_0 is the average value over one DT period.
DT harmonic repetition	$e^{j(k+N_0)\omega_0 n} = e^{jk\omega_0 n}$	Harmonic indices separated by N_0 generate the same DT complex exponential.
DT Fourier-coefficient periodicity	$X_{k+N_0} = X_k$	The DT Fourier-coefficient sequence itself repeats every N_0 indices.
Finite-duration DT Fourier period	$N_0 = B - A + 1$	For a finite DT interval $A, \ldots, B$, the periodic extension has N_0 samples per period.
Finite-duration DT synthesis	$x[n] = \sum_{k=0}^{N_0-1} X_k e^{jk\omega_0 n}, \quad A \le n \le B$	On the original sample interval, the periodic extension's DT Fourier series equals the finite signal.
Finite-duration DT Fourier coefficients	$X_k = \frac{1}{N_0}\sum_{n=A}^{B} x[n] e^{-jk\omega_0 n}$	Computes the N_0 distinct coefficients directly from the finite DT segment.
DT modulation / frequency shifting	$y[n] = e^{jM\omega_0 n} x[n] \Rightarrow Y_k = X_{k-M}$	Multiplication by harmonic M shifts the DT Fourier-coefficient index by M.
DT periodic convolution	$y[n] = \sum_{\ell=0}^{N_0-1} x[\ell] z[n-\ell] \Rightarrow Y_k = N_0 X_k Z_k$	Periodic DT convolution becomes coefficient-wise multiplication, scaled by N_0.
DT multiplication	$y[n] = x[n]z[n] \Rightarrow Y_k = \sum_{\ell=0}^{N_0-1} X_\ell Z_{k-\ell}$	DT multiplication becomes periodic convolution of the Fourier coefficients.
DT conjugation	$y[n] = x^*[n] \Rightarrow Y_k = X^*_{-k}$	Conjugating a DT signal conjugates the coefficient and reverses its index.
DT average power per period	$P_x = \frac{1}{N_0}\sum_{n=0}^{N_0-1} \lvert x[n]\rvert^2$	Average squared magnitude of a periodic DT signal over one period.
DT Parseval relation	$\frac{1}{N_0}\sum_{n=0}^{N_0-1} \lvert x[n]\rvert^2 = \sum_{k=0}^{N_0-1} \lvert X_k\rvert^2$	DT average power equals the sum of the magnitude-squared Fourier coefficients.
Fourier approximation error signal	$e_N(t) = x(t) - x_N(t)$	Difference between the original periodic signal and its truncated Fourier-series approximation.
Approximation error power	$P_{e_N} = \frac{1}{T_0}\int_0^{T_0} \lvert e_N(t)\rvert^2 dt$	Mean-squared approximation error over one period.

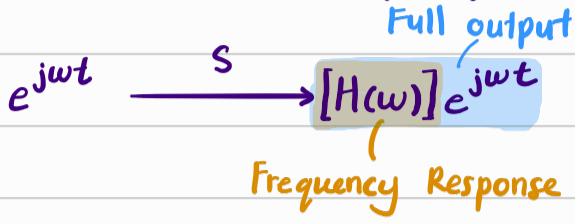
Terminology / notation	Formula	Concise explanation
Error power from omitted Fourier coefficients	$P_{e_N} = \sum_{\lvert k\rvert > N} \lvert X_k\rvert^2$	By Parseval, the approximation error power is the power in the discarded harmonics.
Error power as total minus retained power	$P_{e_N} = \sum_{k=-\infty}^{\infty} \lvert X_k\rvert^2 - \sum_{k=-N}^{N} \lvert X_k\rvert^2$	Equivalent way to compute the power lost by truncating the Fourier series.

WEEK 5

Frequency Response of LTI systems

CT Frequency Response, H(ω)

Previously we learned that impulse response is how the system responds to one time-domain building block ($\delta(t)$). Similarly, frequency response is how the system responds to one frequency-domain building block ($e^{j\omega t}$).



Note that the complex exponential has a very special property when passed through an LTI system: The output is still a complex exponential at the same frequency, only the multiplyer changed.

frequency response

Concept Check

Impulse response, $h(t)$	frequency response, $H(\omega)$
Input : $\delta(t)$	Input : $e^{j\omega t}$
Full output : $h(t)$	Full output : $H(\omega)e^{j\omega t}$
$h(t)$ describes the complete output signal produced by $\delta(t)$	$H(\omega)$ describes the complex scaling applied to $e^{j\omega t}$

What exactly is $H(\omega)$?

For a selected frequency (ω), $H(\omega)$ is generally complex

$$H(\omega) = |H(\omega)|\, e^{j\angle H(\omega)}$$

Magnitude; phase shift given to that frequency

↓ Full output

$$H(\omega)e^{j\omega t} = |H(\omega)|\, e^{j[\omega t + \angle H(\omega)]}$$

Connection btwn impulse respone & frequency response

Using alternate convolution form : $y(t) = \int_{-\infty}^{\infty} h(\tau)\, x(t-\tau)\, d\tau$

Let $x(t) = e^{jwt}$

$$\Longrightarrow y(t) = \int_{-\infty}^{\infty} h(\tau)\left[(e^{jwt})(e^{-jw\tau})\right] d\tau$$

$$= \int_{-\infty}^{\infty} \underbrace{\left[h(\tau)\, e^{-jw\tau}\right]}_{H(w)} e^{jwt}\, d\tau$$

e^{jwt} : constant for a particular t and w

$$H(w) = \int_{-\infty}^{\infty} h(\tau)\, e^{-jw\tau}\, d\tau \quad \text{for all } w$$

Quick Note

When the input is e^{jwt}, the output is $H(w)\, e^{jwt}$

That is a special case.

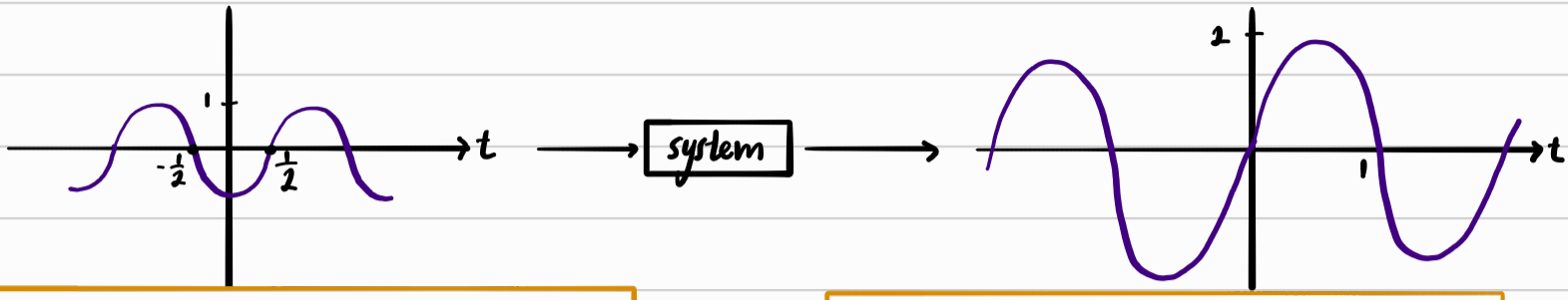
It does not hold true for other arbitrary inputs.

$y(t) = [H(w)]\, x(t)$ is only true when $x(t) = e^{jwt}$.

[For ECE2111 course]

Short Example

Given the following input & output, deduce the frequency response



Amplitude, $A_{in} = 1$

Period, $T = 2 \Rightarrow \omega = \frac{2\pi}{2} = \pi$

Phase shift, $\phi_{in} = -\pi$ for cosine

$\therefore x(t) = \cos(\pi t - \pi)$

Amplitude, $A_{out} = 2$

Period, $T = 2 \Rightarrow \omega = \frac{2\pi}{2} = \pi$

Phase shift, $\phi_{out} = -\frac{\pi}{2}$ for cosine

$\therefore y(t) = 2\cos(\pi t - \frac{\pi}{2})$

$$\therefore |H(\pi)| = \frac{A_{out}}{A_{in}} = \frac{2}{1} = 2 \quad \#$$

Phase change caused by system

$$\therefore \angle H(\pi) = \phi_{out} - \phi_{in}$$

$$= \left[-\frac{\pi}{2}\right] - [-\pi]$$

$$\angle H(\pi) = \frac{\pi}{2} \quad \#$$

This can be evaluated visually too since the output is just faster by 90°.

To explicitely show the $H(\pi)$, the output can be written as follws

$$y(t) = \underbrace{2}_{|H(\pi)|} \underbrace{\cos((\pi t - \pi) + \frac{\pi}{2})}_{x(t)} \quad \angle H(\pi)$$

This example shows that the output of a complex exponential (sinusoid) input is just the "complexly scaled" input itself

$H(\omega)$

Note that we cannot find the entire function $H(\omega)$ cuz that require us to get every possible frequency. From this question, the only values we can deduce are limited since we were only given one input-output pair

For the sake of understanding, let's observe the short example in terms of complex exponentials.

Converting $x(t) = \cos(\pi t - \pi)$ $\xrightarrow{\cos\theta = \frac{1}{2}e^{j\theta} + \frac{1}{2}e^{-j\theta}}$ $x(t) = \frac{1}{2}e^{j[\pi t - \pi]} + \frac{1}{2}e^{-j[\pi t - \pi]}$

$\xrightarrow{e^{\pm j\pi} = -1}$ $x(t) = -\frac{1}{2}e^{j\pi t} - \frac{1}{2}e^{j(-\pi)t}$

So the input contains 2 complex exponentials frequency component

Converting $y(t) = 2\cos(\pi t - \frac{\pi}{2})$ $\xrightarrow{\cos\theta = \frac{1}{2}e^{j\theta} + \frac{1}{2}e^{-j\theta}}$ $y(t) = e^{j[\pi t - \pi/2]} + e^{-j[\pi t - \pi/2]}$

$\xrightarrow{e^{j(\pm\frac{\pi}{2})} = \pm j}$ $y(t) = -je^{j\pi t} + je^{j(-\pi)t}$

For frequency (π)

$$H(\pi) = \frac{\text{Output Complex Coefficient}}{\text{Input Complex Coefficient}}$$

$$= \frac{(-j)}{-1/2}$$

$$= 2j \qquad \angle H(\pi)$$

$$H(\pi) = 2e^{j(\pi/2)}$$

$|H(\pi)|$

For frequency $(-\pi)$

$$H(\pi) = \frac{\text{Output Complex Coefficient}}{\text{Input Complex Coefficient}}$$

$$= \frac{j}{-1/2}$$

$$= 2(-j) \qquad \angle H(-\pi)$$

$$H(-\pi) = 2e^{j(-\pi/2)}$$

$|H(-\pi)|$

$$\therefore \quad x(t) = \left[-\frac{1}{2}e^{j(\pi)t}\right] + \left[-\frac{1}{2}e^{j(-\pi)t}\right]$$

$$y(t) = [H(\pi)]\left[-\frac{1}{2}e^{j(\pi)t}\right] + [H(-\pi)]\left[-\frac{1}{2}e^{j(-\pi)t}\right]$$

Using the normal sinusoid method, we treat the input as one real sinusoid and compare the input & output amplitudes & phases directly.

Using the complex exponential method, the same real sinusoid is split into its w and $-w$ complex frequency components. So the complex-exponential view makes it explicit that the frequency response is frequency-dependent and acts separately on each frequency component.

DT Frequency Response, H(ω)

Same core concept as CT.

For a BIBO-stable, LTI system,

$x[n] = e^{j\omega n}$ ↓ produce $y[n] = H(\omega)\, e^{j\omega n}$

where $H(\omega) = \sum_{n=-\infty}^{\infty} h[n]\, e^{-j\omega n}$

As learned in Fourier Series, the crucial difference is that for DT,

$$e^{j\omega n} = e^{j(\omega + 2\pi)n} = [e^{j\omega n}][e^{j(2\pi)n}]$$

$e^{j(2\pi)n} = 1$ since $n \in \mathbb{Z}$

Therefore ω and $\omega + 2\pi$ are actually the same DT complex exponential.

∴ The system must respond identically:

$$H(\omega + 2\pi) = H(\omega)$$

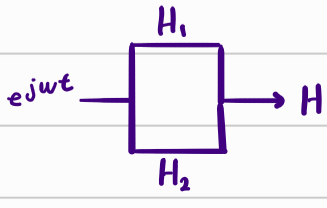
So the DT frequency response itself is periodic in ω with period 2π.

Frequency response interconnected system

① Cascade / Series

$$H_{series}(\omega) = [H_1(\omega)][H_2(\omega)]$$

② Parallel



$$H_{parallel}(\omega) = H_1(\omega) + H_2(\omega)$$

Steady state & Transient Response

Previously, we just defined the input as $x(t) = e^{jwt}$ which means it exists for all time $-\infty < t < \infty$ and thus doesn't rlly have a "start"

However physical signal often turn on at some time and thus having a temporary start-up behaviour caused by switching the input on.

∴ Assuming the input is $x(t) = e^{jwt}u(t)$
The output of a BIBO-stable LTI system becomes

$$y(t) = [H(w)\, e^{jwt}\, u(t)] + y_{tr}(t)$$

Steady-state response — Transient Response

Steady-state response $[H(w)\, e^{jwt}\, u(t)]$

This is the long-term frequency response behaviour

Transient response $[y_{tr}(t)]$

This is the temporary start-up behaviour caused by switching the input on.
Eventually as time goes by, $y_{tr} \to 0$

$$y(t) = H(w)\, e^{jwt}\, u(t) + y_{tr}(t) \xrightarrow{\text{after enough time}} y(t) = H(w)\, e^{jwt}\, u(t)$$

This is why $H(w)$ frequency response describes the steady-state behaviour of the system.

WEEK 6

Filtering

Response of LTI systems to Periodic Signals [Introduction]

Previously, in Week 4 [Fourier Series], we learned that a complicated periodic signal can be represented by a sum of simple complex-exponential frequency components.

$$x(t) = \sum_{k=-\infty}^{\infty} X_k e^{j(k\omega_0)t}$$

$X_k e^{j(k\omega_0)t}$: k-th harmonic with frequency $(k\omega_0)$

Then in Week 5, we learned how an LTI responds to one complex exponential [Frequency Response]

$$e^{j\omega t} \xrightarrow{LTI} H(\omega)\, e^{j\omega t}$$

Week 6

∴ If a periodic signal (week 4) is already made up of many complex-exponentials, we now know how to process each complex-exponential individually. (week 5)

CT Periodic Input through an LTI system

Suppose the periodic input has fundamental ω_0 :

$$x(t) = \sum_{k=-\infty}^{\infty} X_k \, e^{j(k\omega_0)t}$$

From frequency response,

$$e^{j(k\omega_0)t} \xrightarrow{LTI} X_k \, H(k\omega_0) \, e^{j(k\omega_0)t}$$

So each harmonic is treated independently according to the frequency response at that harmonic's own frequency.

∴ Summing all output harmonics back together :

$$y(t) = \sum_{k=-\infty}^{\infty} X_k \, H(k\omega_0) \, e^{j(k\omega_0)t}$$

Fourier Coefficient of the Output

The Fourier Series of the output is

$$y(t) = \sum_{k=-\infty}^{\infty} Y_k \, e^{j(k\omega_0)t}$$

By comparison we get that

Amount of harmonic k in output

Amount of harmonic k in input

$$Y_k = H(k\omega_0) \, X_k$$

System response at that frequency

Take note :

The system does not use one single value of H for the whole signal.

Instead :

$k = 0 \rightarrow H(0)$

$k = 1 \rightarrow H(\omega_0)$

$k = 2 \rightarrow H(2\omega_0)$

⋮

Every frequency component uses the value of the same frequency-response function $H(\omega)$ evaluated at its own frequency.

Magnitude and Phase of Each Harmonic

Since $Y_k = H(k\omega_0) X_k$ and $X_k = |X_k| e^{j \angle X_k}$; $H(k\omega_0) = |H(k\omega_0)| e^{j \angle H(k\omega_0)}$

We get the following:

$|Y_k|$ $\quad$ $\angle Y_k$

$$Y_k = |H(k\omega_0)||X_k| e^{j[\angle H(k\omega_0) + \angle X_k]}$$

$$|Y_k| = |X_k||H(k\omega_0)|$$

Input Magnitude $\quad$ System Gain

$$\angle Y_k = \angle X_k + \angle H(k\omega_0)$$

Input phase $\quad$ System phase shift

So for each harmonic:

$|X_k|$ $\quad$ $|H(k\omega_0)|$

Output magnitude = input magnitude × system gain

$\angle X_k$ $\quad$ $\angle H(k\omega_0)$

Output phase = input phase × system phase

So frequency response lets us understand the output without convolving the entire waveform directly in time.

DT Periodic Input through an LTI system

Same core concept as CT except for the crucial difference that DT only needs N_0 distinct harmonics because the complex exponential repeat in frequency. $[\omega_0 = \frac{2\pi}{N_0}]$

$$y[n] = \sum_{k=0}^{N_0-1} X_k H(k\omega_0) e^{j(k\omega_0)n}$$

$$Y_k = H(k\omega_0) X_k$$

WEEK 8

Fourier Transforms

Fourier Transform Introduction

Previously in Week 4, we learned that the **Fourier Series** gives us a frequency-domain representation of a **periodic signal**. Because a periodic signal repeats exactly every T_0, it can only contain harmonically related frequency components.

This forces the frequency domain to sit on a **discrete harmonic grid**:

$$\dots, -2\omega_0, -\omega_0, 0, \omega_0, 2\omega_0, \dots$$

An aperiodic signal [e.g. single isolated pulse] has no fundamental period. Without T_0, it has no fundamental frequency, ω_0.
If there is no ω_0, there is no "step size" to buid the discrete harmonic grid thus the Fourier Series is no longer the appropriate approx.

To fix this issue, we treat an aperiodic signal as a periodic signal whose period is stretched to infinity [$T_0 \to \infty$]

$\Longrightarrow \omega_0 = \frac{2\pi}{T_0} \to 0$

Meaning to say the discrete harmonic steps packed so closely tgt that the **gaps disappear**, merging into an entire **continuum of possible frequencies**.

Because the frequency grid now is continuous, we can no longer count frequencies using the integer harmonic index, k. Instead, we evaluate the continuous frequency ω.
∴ The frequency representation fundamentally changes:

A discrete sequence of coefficients, $\{X_k\}$

⇓

A continuous mathematical function, $X(\omega)$

This continuous function, $X(\omega)$ is the Fourier Transform.

Discrete-Time Fourier Transform (DTFT)

Recall the DT Fourier Series analysis formula from Week 4

$$X_k = \frac{1}{N_0} \sum_{n=0}^{N_0-1} x[n] e^{-j(k\omega_0)n}$$

comparison term

where essentially for one chosen frequency $k\omega_0$ of harmonic k, $\sum_{n=0}^{N_0-1}$ scans through one complete period of the signal and accumulates the comparison, generating X_k = how strongly harmonic k is present in $x[n]$

As discussed in the introduction, we are dealing w aperiodic signals so 2 things fundamentally change

1) Signal doesn't repeat, so we have to inspect the entire signal
   ↳ $n = 0, \dots, N_0 - 1 \implies n = -\infty, \dots, +\infty$

2) Frequency no longer restricted to integer harmonic frequencies $k\omega_0$
   Instead we directly choose any frequency ω
   ↳ $k\omega_0 \implies \omega$

Therefore the Discrete Time Fourier Transform (DTFT) is:

$$X(\omega) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$

compares the signal against a selected frequency

where $\sum_{n=-\infty}^{\infty} |x[n]| < \infty$

Absolutely Summable

[Ensure $X(\omega)$ converges]

Take Note:

The $1/N_0$ term have vanished from DTFS to DTFT cuz as discussed in the introduction, for an aperiodic signal we stretch the period to infinity, $N_0 \to \infty$. This causes $X(\omega)$ to be "inflated" thus a normalizer of $\frac{1}{2\pi}$ is introduced in the inverse DTFT as discussed in the following pg.

Inverse DTFT [IDTFT]

Recall DTFS synthesis:

$$x[n] = \sum_{k=0}^{N_0-1} X_k e^{j(k\omega_0)n}$$

Essentially we scan across hamonic frequencies, $k\omega_0$ to rebuid the time signal via orthogonality. However, as discussed in the introduction, for an aperiodic signal, the frequency grid is now continuous, covering an infinite continuum of values. Thus the discrete Σ scanner/sum becomes a continuous $\int$ scanner/sum

Therefore the inverse DTFT [DTFT synthesis] is:

$$x[n] = \frac{1}{2\pi} \int_0^{2\pi} X(\omega) e^{j\omega n} d\omega$$

ω - frequency building block

Normalizer

$X(\omega)$: Amount of that building block contained in the signal

Do note that eventho $x[n]$ may be completely aperiodic in time, yet $X(\omega)$ is always periodic in frequency $X(\omega + 2\pi) = X(\omega)$. Recall week 1/4

$$e^{j(\omega+2\pi)n} = e^{j\omega n} e^{j(2\pi)n}$$
$$= e^{j\omega n} [1]$$
$$e^{j(\omega+2\pi)n} = e^{j\omega n}$$

So eventho we have a continuous frequency axis doesn't mean we have infinately many distinct frequencies. So $0 \leq \omega \leq 2\pi$ already contains one complete set of distinct frequencies.

Take note:

For DTFT, the scanner scans across discrete time, n. Thus we used discrete summation, Σ

For Inverse DTFT, the scanner scans across a continuum of frequency, ω. Thus we used Continuous Summation, $\int$

Continuous-Time Fourier Transform (CTFT)

Recall CT Fourier Series Analysis:

$$X_k = \frac{1}{T_0}\int_a^{a+T_0} x(t)\, e^{-j(k\omega_0 t)}\, dt$$

As discussed in the introduction, we are dealing w aperiodic signals so 2 things fundamentally change

1) Signal doesn't repeat, so we have to inspect the entire signal
   ↳ $n = 0, \dots, N_0 - 1 \implies n = -\infty, \dots, +\infty$

2) Frequency no longer restricted to integer harmonic frequencies $k\omega_0$
   Instead we directly choose any frequency ω
   ↳ $k\omega_0 \implies \omega$

Therefore the CTFT is:

$$X(\omega) = \int_{-\infty}^{\infty} x(t)\, e^{-j\omega t}\, dt$$

$e^{-j\omega t}$: compares the signal against a selected frequency

where $\int_{-\infty}^{\infty} |x(t)| < \infty$

Absolutely Integrable

[Ensure $X(\omega)$ converges]

Take Note:

The $1/T_0$ term have vanished from CTFS to CTFT cuz as discussed in the introduction, for an aperiodic signal we stretch the period to infinity, $T_0 \to \infty$. This causes $X(\omega)$ to be "inflated" thus a normalizer of $\frac{1}{2\pi}$ is introduced in the inverse CTFT as discussed in the fllwing pg.

Inverse CTFT [ICTFT]

Recall CT Fourier Series Sythesis:

$$x(t) = \sum_{k=-\infty}^{\infty} X_k e^{j(kw_0)t}$$

As discussed in the introduction, for an aperiodic signal, the frequency grid is now continuous, covering an infinite continuum of values. Thus the discrete scanner/sum (Σ) becomes a continuous scanner/sum ($\int$)

Therefore the CT Fourier Transform is:

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(w)\, e^{jwt}\, dw$$

- w - frequency building block
- $\frac{1}{2\pi}$: Normalizer
- $X(w)$: Amount of that building block contained in the signal

Take Note:

The limits of integration is different for inverse DTFT & CTFT.

DTFT is 2π-periodic in w because discrete-time, n is restricted to integers only. [$e^{j(w+2\pi)n} = e^{jwn}$]

CTFT is generally aperiodic in w because continuous-time, t can take any real values.

A more rigorous proof for the $1/2\pi$

Since $X(\omega)$ is "inflated", we can deflate it by "deflating" $X(\omega)$ by multiplying it back with $\frac{1}{T_0} = \frac{\omega_0}{2\pi} = \frac{\Delta\omega}{2\pi}$

$$x(t) = \sum \left[\frac{\Delta\omega}{2\pi} X(\omega) \right] e^{j\omega t}$$
$$= \frac{1}{2\pi} \sum X(\omega) e^{j\omega t} \Delta\omega$$

As $T_0 \to \infty$, the frequency grid becomes a continuum, $\Delta\omega \to d\omega$ and $\sum \to \int$.

$$\boxed{x(t) = \frac{1}{2\pi} \int X(\omega) e^{j\omega t} \, d\omega}$$

Fourier Transform Properties

Similar to Fourier Series properties, once we know $X(\omega)$, we often don't need to redo the analysis integral everytime $x(t)$ changes. Many time-domain operations corresponds to simple changes in the frequency domain.

① Linearity

If $y(t) = \alpha x_1(t) + \beta x_2(t)$

$$Y(\omega) = \alpha X_1(\omega) + \beta X_2(\omega)$$

② Conjugation

If $y(t) = x^*(t)$ and purely real $x(t) = x^*(t)$

$$X(-\omega) = X^*(\omega)$$

- $|X(-\omega)| = |X(\omega)|$ — Even
- $\angle X(-\omega) = -\angle X(\omega)$ — Odd

③ Time Shifting

If $y[n] = x[n-N]$

$$Y(\omega) = e^{-j\omega N} X(\omega)$$

If $y(t) = x(t-\tau)$

$$Y(\omega) = e^{-j\omega\tau} X(\omega)$$

Time shift $\longleftrightarrow$ Frequency-dependent phase shift

④ Modulation / Frequency-shift

(Multiplying)

If $y(t) = e^{jw_c t}\, x(t)$

$$Y(w) = X(w - w_c)$$

Multiply by complex exponential in time $\longleftrightarrow$ shift the entire spectrum

⑤ CT time scalling

If $y(t) = x(\alpha t)$

Recall time dilation from Week 1

$$Y(w) = \frac{1}{|\alpha|} X\left(\frac{w}{\alpha}\right)$$

$|\alpha| > 1$: Compress in time $\longleftrightarrow$ Expand in frequency

$0 < |\alpha| < 1$: Expand in time $\longleftrightarrow$ Compress in frequency

Take note:

This generally does not work for DT unless the argument for the Fourier transform, $\frac{w}{\alpha}$ is an integer

⑥ Convolution

If $y = x_1 * x_2$

$$Y(\omega) = X_1(\omega)\, X_2(\omega)$$

Convolution in time ⟷ Multiplication in frequency

⑦ Multiplication in time

If $y(t) = x_1(t)\, x_2(t)$

CT: $$Y(\omega) = x_1 * x_2 = \frac{1}{2\pi}\int_{-\infty}^{\infty} X_1(\Omega)\, X_2(\omega-\Omega)\, d\Omega$$

DT: $$Y(\omega) = x_1 * x_2 = \frac{1}{2\pi}\int_{0}^{2\pi} X_1(\Omega)\, X_2(\omega-\Omega)\, d\Omega$$

Parseval's Relation

Previously in Fourier Series we had a periodic signal. Since a periodic signal continue forever, its total energy is generally infinite. Therefore Parseval was written in terms of average power.

Now Fourier Transform is mainly describing aperiodic signals that generally don't continue forever. So Parseval now is written in terms of total energy.

DT: $$E_{total} = \sum_{n=-\infty}^{\infty} |x[n]|^2 = \frac{1}{2\pi}\int_{0}^{2\pi} |X(\omega)|^2\, d\omega$$

CT: $$E_{total} = \int_{-\infty}^{\infty} |x(t)|^2\, dt = \frac{1}{2\pi}\int_{-\infty}^{\infty} |X(\omega)|^2\, d\omega$$

Energy spread through time = Energy spread through frequency

Frequence Response of LTI systems

Previously in Week 5 we learned that if we had a periodic input:

$$x(t) = \sum_k X_k e^{j(kw_0)t}$$

We learned that an LTI system processes each harmonic independently:

$$e^{j(kw_0)t} \longrightarrow H(kw_0) e^{j(kw_0)t}$$

Therefore,

$$Y_k = H(kw_0) X_k$$

Fourier Transform extends this idea to aperiodic signals.

Given an aperiodic input:

$$x(t) = \frac{1}{2\pi} \int X(w) e^{jwt} dw$$

We learned that an LTI system processes each harmonic independently:

$$e^{jwt} \longrightarrow H(w) e^{jwt}$$

Therefore,

$$y(t) = \frac{1}{2\pi} \int X(w) H(w) e^{jwt} dw$$

By comparison with inverse Fourier Transform:

$$y(t) = \frac{1}{2\pi} \int Y(w) e^{jwt} dw$$

We get:

$$Y(w) = H(w) X(w)$$

Quick Table Summary

Periodic Input	General / Aperiodic input
X_k	$X(w)$
$H(kw_0)$	$H(w)$
$Y_k = H(kw_0) X_k$	$Y(w) = H(w) X(w)$

Connection btwn impulse response & frequency response

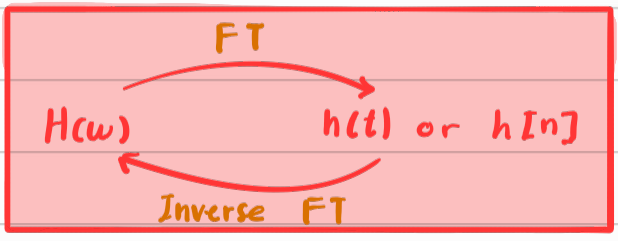
Let $y = x * h$,

Fourier Transform of impulse response

$$Y(\omega) = X(\omega)\,\mathcal{F}\{h\}$$

Comparing to $Y(\omega) = X(\omega)\,H(\omega)$,

$$H(\omega) = \mathcal{F}\{h\}$$



Another way to proove this relationship is via convolution that we did in Week 6.

Connection btwn impulse respone & frequency response

Using alternate convolution form : $y(t) = \int_{-\infty}^{\infty} h(\tau)\,x(t-\tau)\,d\tau$

Let $x(t) = e^{j\omega t}$

$$\Longrightarrow y(t) = \int_{-\infty}^{\infty} h(\tau)\left[(e^{j\omega t})(e^{-j\omega\tau})\right] d\tau$$

$$= \int_{-\infty}^{\infty} \underbrace{\left[h(\tau)\,e^{-j\omega\tau}\right]}_{H(\omega)} e^{j\omega t}\, d\tau$$

($e^{j\omega t}$: constant for a particular t and ω)

$$H(\omega) = \int_{-\infty}^{\infty} h(\tau)\,e^{-j\omega\tau}\,d\tau \quad \text{for all } \omega$$

Comparing this to the CTFT analysis formula,

$$X(\omega) = \int_{-\infty}^{\infty} x(t)\,e^{-j\omega t}\,dt$$

gives that $x(t) = h(t)$, $X(\omega) = H(\omega)$

Meaning that performing CTFT analysis on a impulse response input gives us the frequency response. Therefore also giving

$$H(\omega) = \mathcal{F}\{h\}$$

Low Pass Filter Walkthrough

We just learned that the impulse response & frequency response are 2 description of the same LTI system

$$h(t) \underset{IFT}{\overset{FT}{\rightleftharpoons}} H(\omega)$$

Therefore, if we specify how we want a system to behave in the frequency domain, we can find the corresponding time-domain system by taking the Inverse Fourier Transform.

For an ideal low-pass filter, we define the frequency respose as fllws:

$$H_{LP}(\omega) = \begin{cases} 1, & |\omega| \le \omega_c \\ 0, & |\omega| > \omega_c \end{cases}$$

where ω_c = cut-off frequency

Now to find what time-domain system (impulse response) produces this response, we use:

$$h_{LP}(t) = \mathcal{F}^{-1}\{H_{LP}(\omega)\}$$
$$= \frac{1}{2\pi}\int_{-\infty}^{\infty} H_{LP}(\omega)\, e^{j\omega t}\, d\omega$$

Since $H_{LP} = 0$ outside $-\omega_c \le \omega \le \omega_c$ and $H_{LP} = 1$ inside $-\omega_c \le \omega \le \omega_c$

$$\Longrightarrow h_{LP} = \frac{1}{2\pi}\int_{-\omega_c}^{\omega_c} e^{j\omega t}\, d\omega$$
$$= \frac{1}{2\pi j t}\left[e^{j\omega_c t} - e^{-j\omega_c t}\right]$$
$$= \frac{1}{2\pi j t}\left[2j \sin(\omega_c t)\right]$$
$$h_{LP}(t) = \frac{\sin(\omega_c t)}{\pi t} \quad \text{CT}$$

OR

$$h_{LP}[n] = \frac{\sin(\omega_c n)}{\pi n} \quad \text{DT}$$

The main point here is that

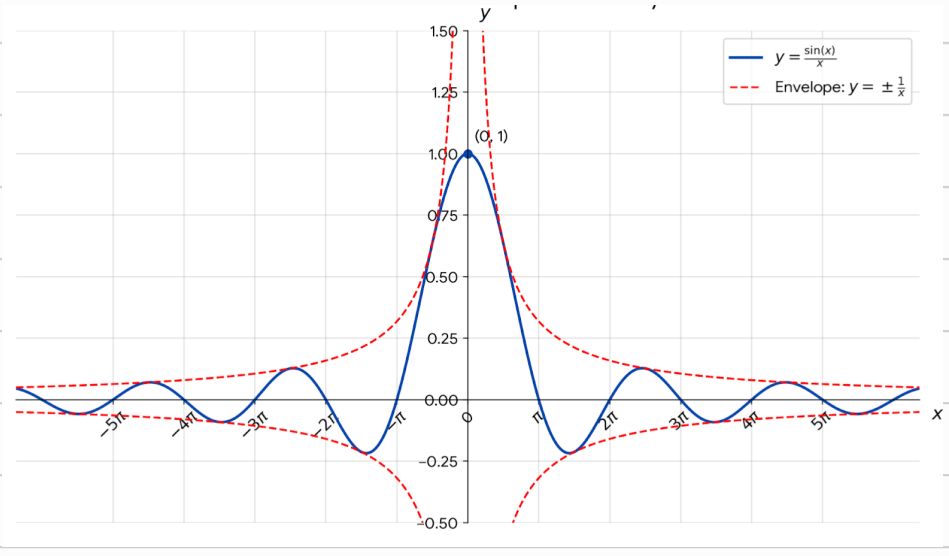
Rectangular $H_{LP}(\omega)$ $\longleftrightarrow$ sinc-shaped h_{LP}

(Explained in next page)

Sinc Function, sinc(t)

Because functions similar to $\frac{\sin(\omega_c t)}{\pi t}$ appear often in Fourier Transform, we define the sinc function

$$\text{sinc}(t) = \begin{cases} \frac{\sin(t)}{t}, & t \neq 0 \\ 1, & t = 0 \end{cases}$$



The sinc function is essentially a sin function but with

1) Decaying amplitude/oscillation of factor $\frac{1}{|t|}$
2) Peaks at 1 as the origin
3) Slightly shifted peaks

Refering back to our LPF example,

$$h(t) = \frac{\sin(\omega_c t)}{\pi t} = \frac{\omega_c}{\pi}\,\text{sinc}(\omega_c t)$$

A sinc function is NOT causal

Recall from Week 3 that the impulse response $h(t)$ is the complete output produced by a single unit impulse applied exactly at time zero ($t = 0$).

AND

A causal system may use the current or/and past inputs, but a future input cannot affect the output now.

∴ Since $h(t) = \text{sinc}(t) \neq 0$ for $t < 0$

↳ sinc(t) is NOT causal

Also notice that when the frequency response $H(\omega)$ was very tightly confined, its corresponding impulse response $h(t)$ actually extends indefinitely through time.
Therefore we have the following relationship:

> More localised in frequency ⟷ Less localized in time

> More localised in time ⟷ Less localised in frequency

Generalised Fourier Transform [Introduction]

Previously we learned 2 different ways to represent signals:

① Fourier Series ~ Periodic Signals $x(t) = \sum_{k=-\infty}^{\infty} X_k e^{j(k\omega_0)t}$

↳ Gives a discrete frequency representation [Week 4, Pg 32]

② Fourier Transform ~ Aperiodic Signals $x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega$

↳ Gives a continuous frequency representation

In real-world systems, periodic & aperiodic signals interact all the time
However to find the output, we want to avoid using 2 separate rules:

Periodic : $Y_k = H(k\omega_0) X_k$ [FS]

Aperiodic : $Y(\omega) = H(\omega) X(\omega)$ [FT]

rather we express both periodic & aperiodic signals using the same Fourier Transform viewpoint and just use one rule: $Y(\omega) = H(\omega) X(\omega)$.

∴ The generalised Fourier Transform allows us to describe both using the same frequency-domain idea.

Generalised Fourier Transform

The problem FT have with periodic signals is that a periodic signal is NOT absolutely integrable cuz it repeats forever.
This is a problem cuz as learned previously, for the ordinary CTFT to work, a signal must be absolutely integrable

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} \, dt$$

compares the signal against a selected frequency

where $\int_{-\infty}^{\infty} |x(t)| < \infty$

Absolutely Integrable

[Ensure $X(\omega)$ converges]

However, Fourier Series already tells us that periodic signals contains discrete frequencies. So we essentially "bruteforce" these discrete frequencies into the FT graph by manually pasting the discrete FS results onto the continuous FT graph.

To do this, we use the Dirac delta impulse (δ). By planting a Dirac Delta at a harmonic frequency, we manually declared a massive conc. of energy at that discrete frequency.
Therefore instead of saying:

$X(\omega)$ = Continuous frequency distribution

We allow:

$X(\omega)$ = Continuous frequency distribution + Frequency Impulse

From Aperiodic Signal — From Periodic Signal

Complex Exponential Result

Let's try this with the simplest periodic signal:

$$x(t) = e^{j\omega_0 t}$$

From FS, the signal only contains one frequency:

$$\omega_0$$

So we put a Dirac impulse frequency at ω_0:

$$\delta(\omega - \omega_0)$$

Therefore:

$$e^{j\omega_0 t} = 2\pi\, \delta(\omega - \omega_0)$$

This is to cancel off the $\frac{1}{2\pi}$ in inverse FS

Check:

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega)\, e^{j\omega t}\, d\omega$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} [2\pi\, \delta(\omega - \omega_0)]\, e^{j\omega t}\, d\omega$$

Since impulse select $\omega = \omega_0$

$$\Rightarrow x(t) = e^{j\omega_0 t}$$

Connecting FS with generalised FT

With the complex exponential result, we can now construct the generalised FT.

Recall FS : $x(t) = \sum_{k=-\infty}^{\infty} X_k e^{j(k\omega_0)t}$

where each term $X_k e^{j(k\omega_0)t}$ is one complexly scaled harmonic with harmonic $k\omega_0$

Using the exponential result : $e^{j(k\omega_0)t} = 2\pi \delta(\omega - k\omega_0)$

$\Longrightarrow$ $X_k e^{j(k\omega_0)t} = 2\pi X_k \delta(\omega - k\omega_0)$

Adding every harmonic then gives us :

Generalized FT $\boxed{X(\omega) = \sum_{k=-\infty}^{\infty} 2\pi X_k \delta(\omega - k\omega_0)}$

Take Note : Terminology

Ordinary FT : Only for aperiodic signals

Generalised FT : Only for periodic signals

If a mixed signal was given, by linearity

FT $X(\omega)$ = (Ordinary FT) + (Generalised FT)

Clean Summary Table

Signal	Main tool	Frequency representation
CT periodic	CTFS	X_k: discrete harmonics
DT periodic	DTFS	X_k: N_0 distinct harmonics
CT aperiodic	CTFT	$X(\omega)$: continuous, generally aperiodic
DT aperiodic	DTFT	$X(\omega)$: continuous, 2π-periodic

Generalised FT allow periodic signals to also be written on the FT graph (Continuous ω-axis)

WEEK 9

Sampling & Reconstruction

Sampling

Back in Week 1, we defined sampling by taking a CT signal, $x(t)$ and recording its values every T_s seconds:

$$x_s[n] = x(nT_s)$$

- n-th sample
- T_s: Sampling period (seconds/sample)

$$f_s = \frac{1}{T_s}$$

- f_s: Sampling frequency [samples/second]

As an example, suppose

$$x(t) = \cos(\omega_0 t)$$

- ω_0: CT angular frequency (rad/s)
- t: CT time

After sampling, substitute $t = nT_s$:

$$x_s[n] = \cos(\omega_0 T_s n)$$

- $\omega_0 T_s$: DT angular frequency (rad/sample)
- nT_s: CT time of sample-n

Take note:

The following notations might be used to avoid confusion

- ω = CT angular frequency [rad/s]
- $\Omega = \omega T_s$ = DT angular frequency [rad/sample]

There are a few problems with this sampling method as discussed in the next few pages.

Aliasing [Loss of distinguishbility]

Now recall that DT frequency repeats every 2π :

$$e^{j(\omega+2\pi)n} = e^{j\omega n}$$

That means 2 different CT frequencies can become the same DT frequency after sampling.

∴ Since DT frequency repeats every 2π, any CT frequency satisfying

$$\omega_\ell T_s = \omega_0 T_s + (2\pi)\ell \quad , \text{where } \ell \in \mathbb{Z} \quad -①$$

will produce the same sampled DT signal. This is aliasing

Therefore, using ①

Sampling Angular frequency, ω_s

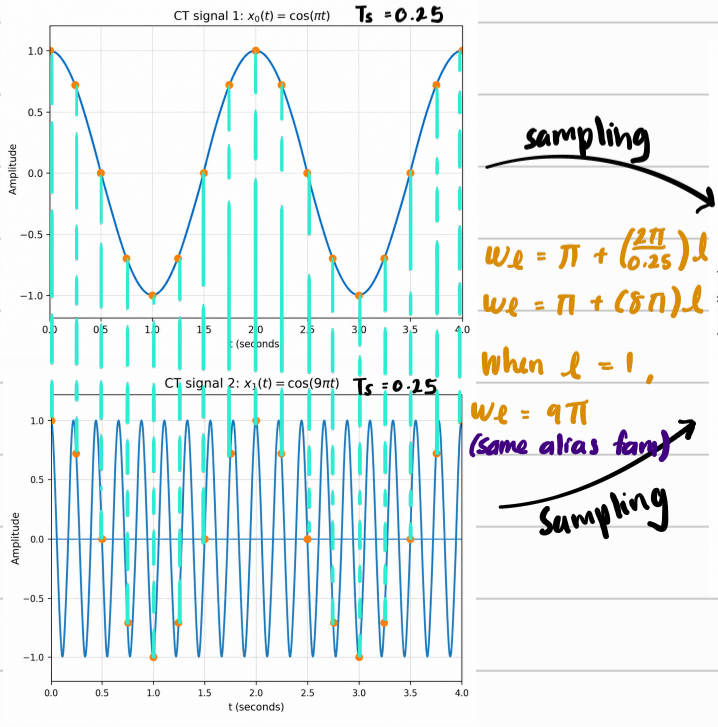
CT angular frequency in the same alias family

$$\omega_\ell = \omega_0 + \left(\frac{2\pi}{T_s}\right)\ell \qquad \text{where } \ell \in \mathbb{Z}$$

$$\omega_\ell = \omega_0 + \omega_s \ell$$

Reference Angular frequency

All CT frequencies that satisfy this equation are from the same alias family. Meaning to say, they all become indistinguishable after sampling.



sampling

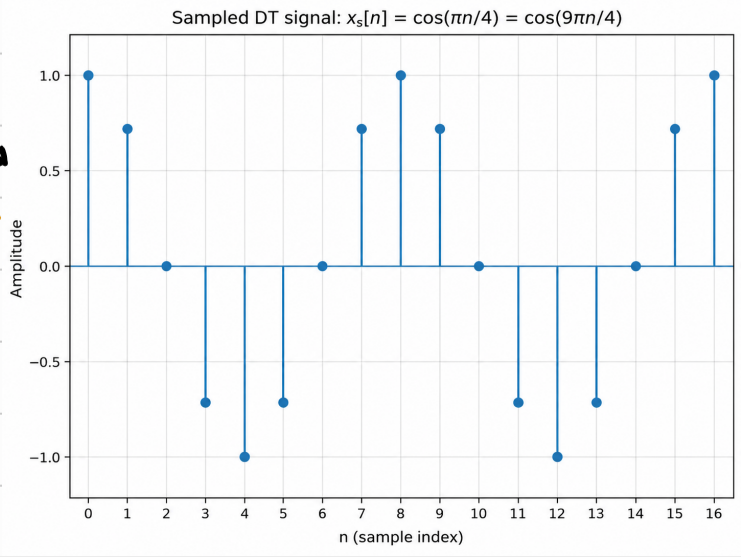
$\omega_\ell = \pi + \left(\frac{2\pi}{0.25}\right)\ell$

$\omega_\ell = \pi + (8\pi)\ell$

When $\ell = 1$,

$\omega_\ell = 9\pi$ (same alias fam)

Sampling



How to avoid Aliasing ?

We already have

$$w_\ell = w_0 + \ell w_s \quad , \text{where } w_s = \frac{2\pi}{T_s}$$

In other words, the alias-family picture is :

$$w_0 \; , \; w_0 + w_s \; , \; w_0 + 2w_s \; , \; \ldots$$

all become indistinguishable after sampling.

To reduce the chance of aliasing, we essentially need to space the alias family further apart. We do this by increasing w_s. Since $w_s = \frac{2\pi}{T_s}$, we simply increase the sampling rate to achieve this

Decrease T_s

Nyquist Frequency

Suppose the original CT sinusoid has frequency w_0

After sampling, the sampled DT sinusoid has frequency $\Omega_0 = w_0 T_s$

And since DT frequencies repeat every 2π, distinct frequencies only exist within the range $-\pi < \Omega < \pi$

$|\Omega| < \pi$

$\therefore$ To keep our CT sinusoid inside that unaliased DT range :

$$|w_0 T_s| < \pi$$

In ECE 2111, $T_s > 0$, $w_0 > 0$

$$w_0 T_s < \pi$$

$$T_s < \frac{\pi}{w_0}$$

$$w_0 < \frac{\pi}{T_s}$$

$$w_0 < \frac{w_s}{2}$$

Nyquist frequency $w_N = \frac{\pi}{T_s} = \frac{w_s}{2}$

$\therefore$ For a sampler operating at w_s, the halfway point $\left(\frac{w_s}{2}\right)$ is the Nyquist frequency (w_N) which marks the absolute highest CT frequency we can sample without aliasing.

DTFT of a Sampled Signal

We already know how sampling affects one frequency (ω_0). But a real CT signal is made of an entire frequency spectrum $X(\omega)$ (CTFT, Week 8)

Let $X(\omega)$ be the original continuous spectrum, and $X_s(\Omega)$ be the new sampled discrete spectrum.

$$X_s(\Omega) = \frac{1}{T_s} \sum_{\ell=-\infty}^{\infty} X\left(\frac{\Omega - 2\pi\ell}{T_s}\right)$$

- $X_s(\Omega)$: DTFT of sampled signal
- $\frac{1}{T_s}$: Amplitude Scaler
- $\sum_{\ell=-\infty}^{\infty}$: Put all clones on the same graph
- $2\pi\ell$: 2π periodic cloner ← Cuz DT must repeat every 2π.
- $\frac{\Omega}{T_s}$: Translator [$\omega = \Omega/T_s$]
- Shifted & Rescaled axis

Essentially, this formula plots the new discrete spectrum (DTFT) by looking up the corresponding data points on the original continuous spectrum (CTFT) $X_s(\Omega) = X\left(\frac{\Omega}{T_s}\right)$.

However DT must be 2π-periodic, so the rest of the formula exist purely to force an originally aperiodic CT spectrum into the periodic frequency structure required by a DT signal.

This begs the question of

> What happens if the clones overlap with one another, wouldn't summing them up ruin the original signal?

this is what we will be discussing about next.

Before we can look at the whole signal overlapping, we need to define how "wide" / "broad" the original signal is.

Bandwidth, W

A CT time signal has a bandwidth of W if all of its frequency content sits strictly btwn $-W$ and W.
In other words,

$$X(\omega) = 0 \quad \text{for} \quad |\omega| > W$$

Aliasing in the Entire Spectrum

From our sampled DTFT formula, we know sampling takes the CT spectrum, maps it to the discrete Ω axis, and clones it every 2π.

Original CT : $-W$ [Spectrum / Content] W

Sampled DT : ... 2π [Copy, $l=-1$] 2π [Copy, $l=0$] 2π [Copy, $l=1$] 2π ...

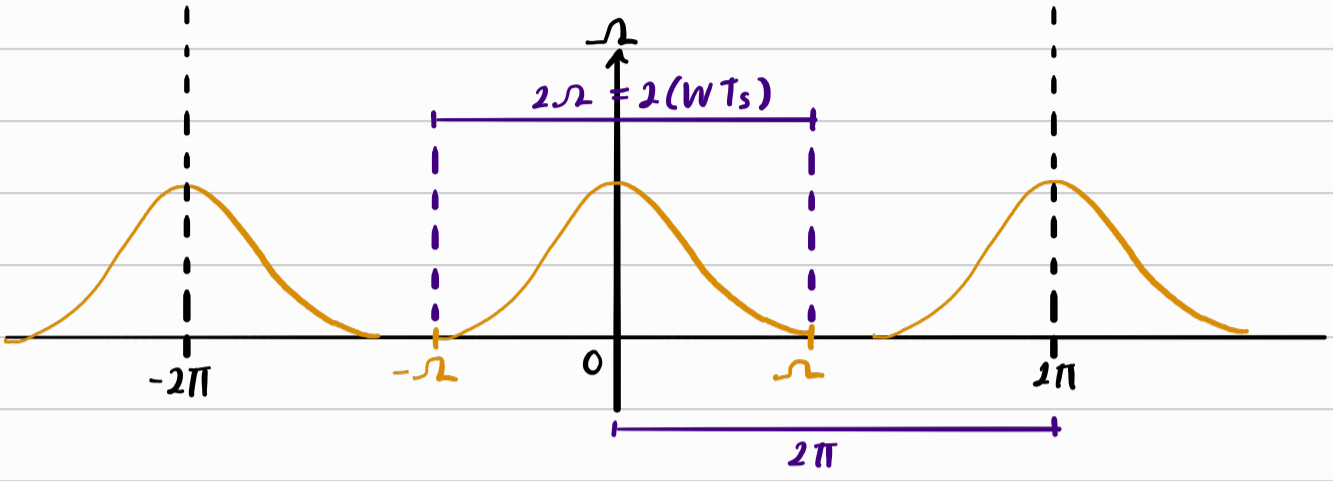
where each [Copy] is : $-\Omega$ [Spectrum / Content] Ω

The sampled DTFT will NOT face aliasing if the copies / clones are "narrow" enough to stay separated. If the copies overlap, because sampled DTFT adds $\sum_l$ the overlapping parts tgt, the original shape is permanently distorted and we can no longer tell if a frequency was originally there, or if it "folded in" from a neighbouring clone.

$\therefore$ Overlapping clones $\Rightarrow$ Loss of distinguishbility (Aliasing)

Shannon - Nyquist Condition (Sampling Theorem)

To avoid aliasing for the entire spectrum, the total width of a single clone must be smaller than the 2π gap btwn the clone centers.



$\therefore$ The no overlap condition : $2(WT_s) < 2\pi$

$$T_s < \frac{\pi}{W}$$

$$w_s = 2\pi/T_s$$

Sampling Theorem $\quad w_s > 2W$

Essentially, to completely avoid aliasing, the sampling rate (w_s) must be strictly greater than twice the bandwidth (W) of the original signal

In ECE 2111, we almost always deal with baseband signals, meaning that the frequency spectrum is perfectly centered around zero ($\omega = 0$)

↳ Highest frequency the signal reaches b4 dropping to zero is exactly W

↳ $f_{highest} = W$

$\therefore$ We can also say to completely avoid aliasing, the sampling rate (w_s) must be strictly greater than twice the frequency of the original signal

$f_{highest} = W$

Nyquist Frequency & Nyquist Rate

Feature	Nyquist Frequency (ω_N)	Nyquist Rate
Belongs to...	The Sampler	The Signal
The Formula	$\omega_N = \frac{\omega_s}{2}$	$\text{Rate} = 2W$
The Question it Answers	"My sampler runs at ω_s. What is the **absolute highest frequency** I can safely record?"	"My signal has a bandwidth of W. What is the **absolute slowest speed** I must sample it at?"
Concept	A speed limit imposed by your hardware.	A hardware requirement demanded by your signal.

Essentially :

To successfully sample a signal without aliasing, the sampler's sampling frequency (ω_s) must be greater than the signal's Nyquist rate ($2W$). Equivalently, the sampler's Nyquist frequency (ω_N) must be greater than the signal bandwidth (W).

Anti - Aliasing Filter

So far we assumed signals were perfectly bandlimited [$X(\omega) = 0$ for $|\omega| > W$].
But in reality, physical measurements pick up unwanted garbage :

Measured Signal = Useful Low-freq. info + High-freq. noise

Bandwidth, W

If we sample this directly, the high-frequency noise will alias (fold) down into the low frequency range. Once the noise folds into the useful DT frequencies, it is permanently mixed with the wanted data.

To fix this, we "destroy" the high frequencies before they even reach the sampler. We do this by placing a Low Pass Filter (LPF) right infront of the sampler.

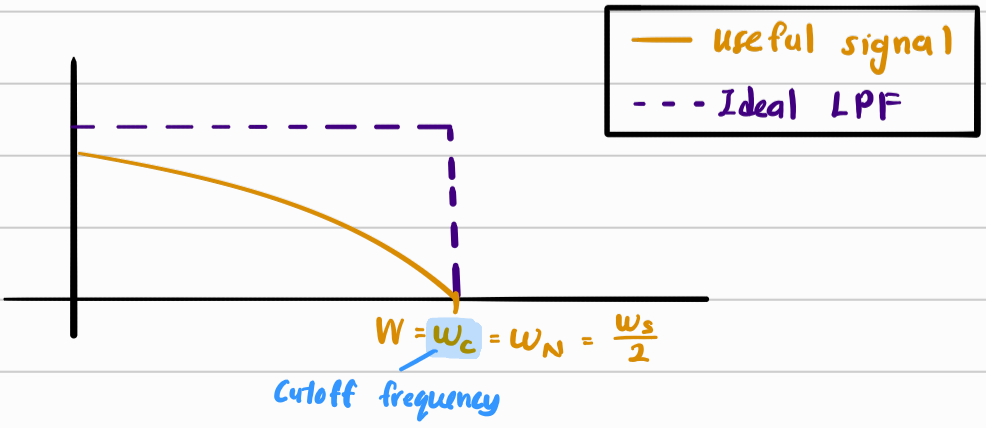
$x(t)$ → Anti-aliasing LPF → Sampler → $x_s[n]$

Anti - Aliasing Filter Design

① Ideal Case

Assuming an Ideal LPF drops instantly from 100% pass to 100% b
Because the ideal LPF have no "roll off" / transient region, we can lock. just set the Nyquist frequency (ω_N) to be right at the end of our useful data (W)

$$\omega_c = \omega_N = \frac{\omega_s}{2} = \frac{\pi}{T_s}$$



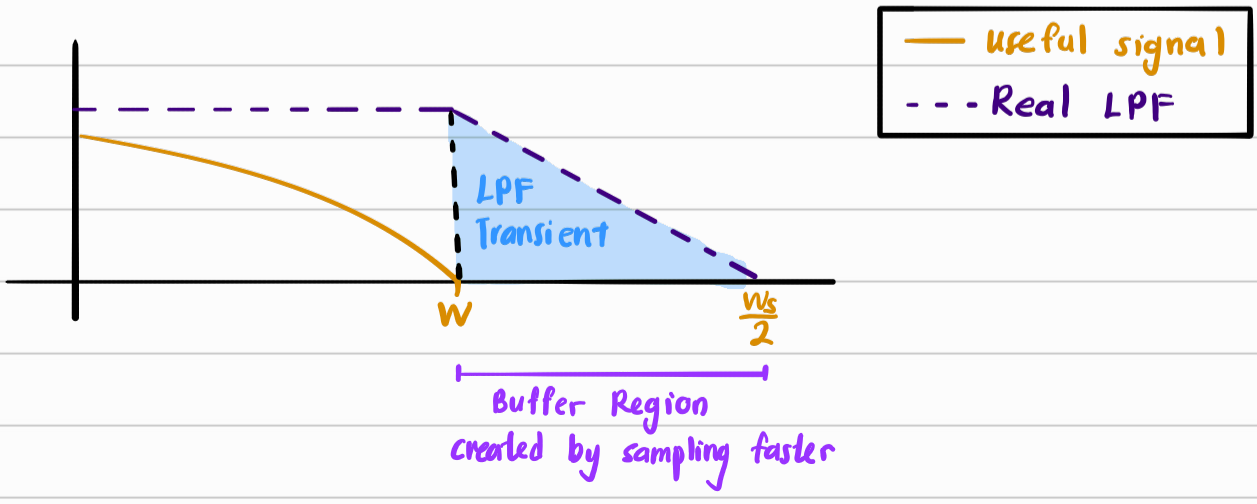
② Practical Case

Real LPF cannot just "turn off" instantly. They slope downward, requiring a transition region to roll off. If we try to force a real LPF to hit zero exactly at W, it will start sloping down early, destroying the top end of our useful data.

∴ To protect our useful data, we must let the filter stay perfectly flat until W, and then let it roll off.
But the roll off must hit zero before the Nyquist limit ($^{W_s}/_2$), or else the leftover noise will alias and fold into the useful data

So we sample faster than 2W to push the Nyquist frequency ($^{W_s}/_2$) further to the right. This creates a Buffer Zone where the filter can safely roll off.

$$W < \text{Transition Region} < \frac{W_s}{2}$$



Signal Reconstruction (DT → CT)

Sampling was converting CT to DT. Reconstruction does the reverse.

$$x_s[n] \longrightarrow y(t)$$

The problem with this is $x_s[n]$ is a discrete sequence, but $y(t)$ is a continuous wave. We can't directly put discrete numbers into a continuous filter.

Step ① : Representing the DT samples into CT format

First, we must translate the discrete samples into a valid CT format. We do this by placing a CT unit impulse (δ) at every sampling time slot, scaled by the sample's value [Refer Week 3]

$$w(t) = \sum_{n=-\infty}^{\infty} x_s[n]\,\delta(t - nT_s)$$

- $\sum_{n=-\infty}^{\infty}$: Put all on same graph
- $x_s[n]$: Scale impulse by actual value
- $\delta(t - nT_s)$: Turn on at sampling time slot (nT_s)

Step ② : Filling the gaps via convolution

Now that $w(t)$ is a valid CT signal, we pass it through a reconstruction filter. This is an LTI system with impulse response $h(t)$ designed to fill up the empty space btwn the spikes. From Week 3, we know the output is just convolution. [$y(t) = w(t) * h(t)$]

$$y(t) = \sum_{n=-\infty}^{\infty} x_s[n]\,h(t - nT_s)$$

- $x_s[n]$: Current scanned sample value
- $h(t)$ centered on sample time slot
- nT_s: Sample Time

Essentially, you take a wide, smooth pulse $h(t)$ and center a copy of it on every single sample time (nT_s), then scale it by the sample value $x_s[n]$. Then you add all these overlapping pulses tgt to create a continuous wave, $y(t)$.

Step ③ : The Interpolation Condition

The problem with step ② is that if there are overlapping "tails" from neighbouring pulses, it will corrupt our original sample values.
To be a true reconstruction, the final continuous wave $y(t)$ must pass exactly through all sample points.

$$y(mT_s) = x_s[m] \quad \Longrightarrow \quad y(mT_s) = \sum_{n=-\infty}^{\infty} x_s[n]\, h((m-n)T_s)$$

target sample location

scanning thru every sample

Since we want the output to equal exactly $x_s[m]$, in other words we want all contribution to vanish except the sample $n = m$.
So the reconstruction pulse $h(t)$ must equal 1 at its centre, zero everywhere else.

⟹ Mathematically, the filter must satisfy the Interpolation Condition:

$$h(kT_s) = \delta[k]$$

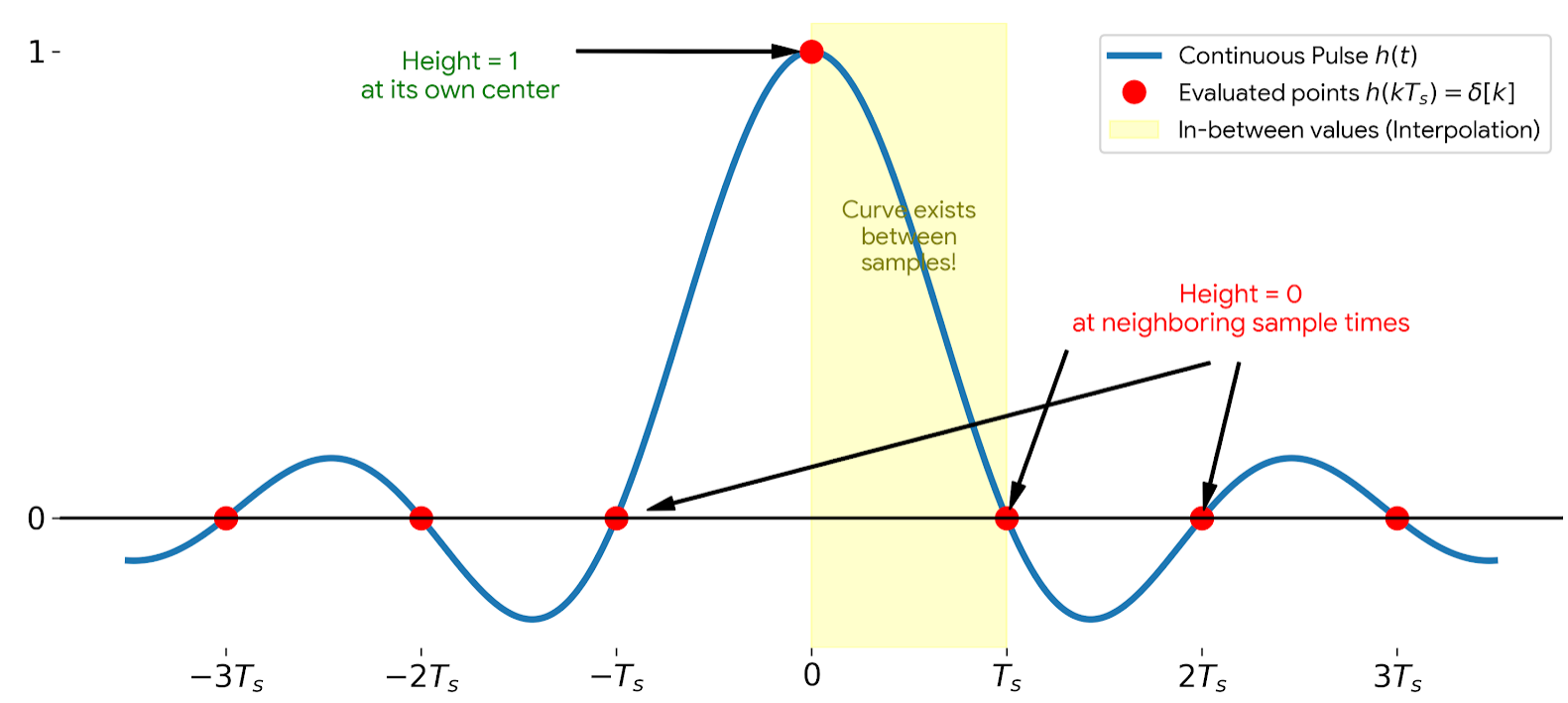
meaning : $h(0) = 1 \;;\; h(1T_s) = h(12T_s) = \ldots = 0$

∴ With the interpolation condition in place, the condition of the final continuous wave $y(t)$ passing through all sample points is satisfied.
⟹ At $t = mT_s$, only the $n = m$ term remains.

$$y(mT_s) = x_s[m] \checkmark$$

Essentially reconstruction works by replacing every discrete dot with a continuous wave $h(t)$. As long as this wave is shaped so that it perfectly crosses the zero-axis as it passes every other sample's time slot, its overlapping tails will fill the empty space btwn samples without accidentally corrupting the original known data points.

[Illustration in next pg]


The Interpolation Condition: $h(kT_s) = \delta[k]$
1
0
Height = 1
at its own center
Curve exists
between
samples!
Height = 0
at neighboring sample times
Continuous Pulse $h(t)$
Evaluated points $h(kT_s) = \delta[k]$
In-between values (Interpolation)
$-3T_s$
$-2T_s$
$-T_s$
0
T_s
$2T_s$
$3T_s$

Shape of reconstruction pulse, $h(t)$

There are many different $h(t)$ that satisfy the interpolation condition.

So the natural next question is:

What $h(t)$ shud we use to fill the gaps?

① Zero-Order Hold (ZOH)

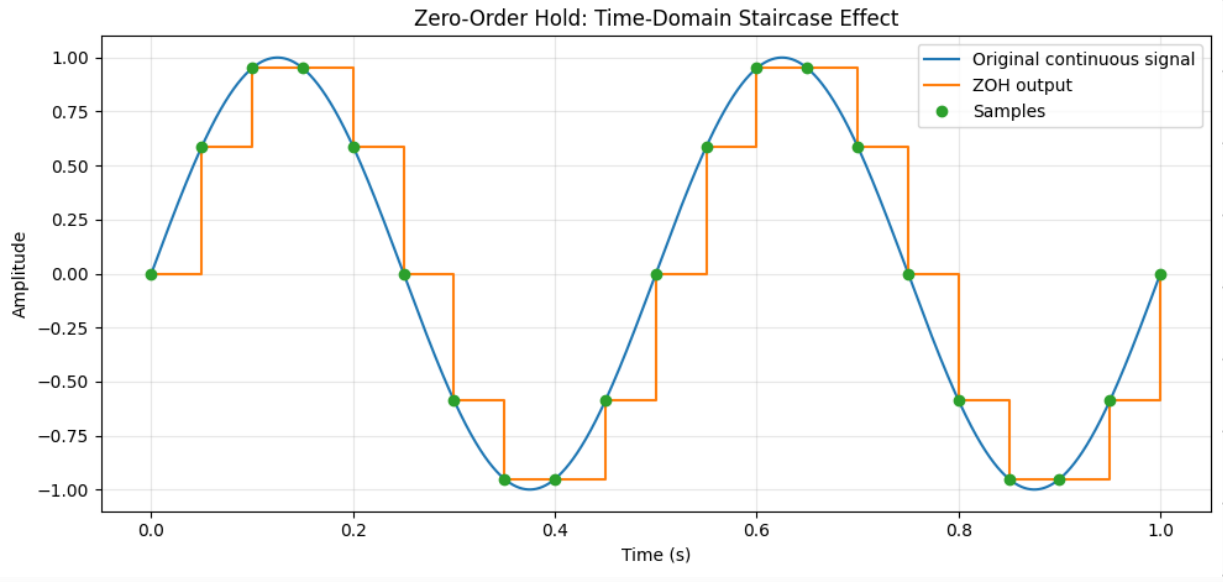
The simplest practical circuit that just grabs a sample's value and hold it completely flat until the next sample arrives.

$$h_{ZOH}(t) = u(t) - u(t - T_s)$$

$$h_{ZOH}(t) = 1, \quad 0 \le t \le T_s$$

So the general reconstruction formula becomes

$$y(t) = \sum_{n=-\infty}^{\infty} x_s[n]\, h_{ZOH}(t - nT_s)$$



Essentially, instead of joining the discrete dots smoothly, ZOH turns each dot into a flat rectangular block lasting T_s.

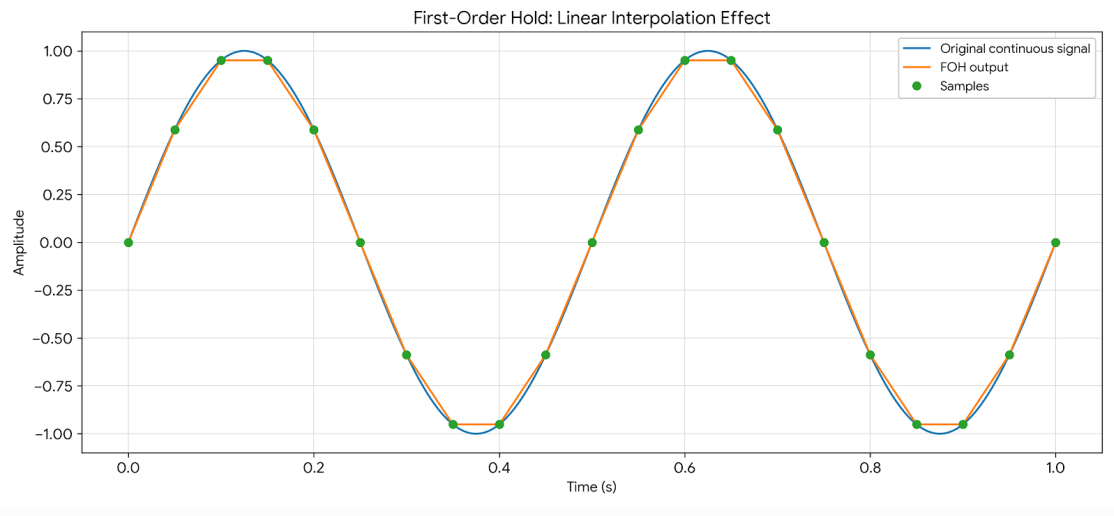
② First-Order Hold (FOH)

Instead of leaving the gaps flat, a slightly smarter circuit calculates the slope and draws a straight line directly to the next sample point [Linear Interpolation]

$$h_{FOH}(t) = \begin{cases} 1 + t/T_s, & -T_s \le t \le 0 \\ 1 - t/T_s, & 0 \le t \le T_s \\ 0, & \text{otherwise} \end{cases}$$

Notice that $h_{FOH}(0) = 1$ and $h_{FOH}(\pm T_s) = h_{FOH}(\pm 2T_s) = \dots = 0$

$\Longrightarrow$ Satisfies the interpolation condition of $h_{FOH}(kT_s) = \delta[k]$



To summarize, the general reconstruction mechanism never changed

$$y(t) = \sum_n x_s[n]\, h(t - nT_s)$$

The only thing that changed is the reconstruction pulse, $h(t)$

- ZOH ~ Fill gaps with horizontal lines (Staircase)
- FOH ~ Fill gaps with straight sloped lines (Linear Interpolation)

Frequency - Domain View of Reconstruction

We just built the physical time-domain engine for reconstruction. Now using our Week 8 Fourier rules, we can map this exact same process into the frequency-domain.

Step ① : Representing the DT samples into CT format [Frequency - Domain]

Recall our CT impulse train :

$$w(t) = \sum_n x_s[n]\,\delta(t - nT_s)$$

CTFT ↓

$$W(\omega) = \sum_{n=-\infty}^{\infty} x_s[n]\, e^{-j\omega n T_s}$$

DTFT of our discrete samples, $x_s[n]$

$$X(\Omega) = \sum_{n=-\infty}^{\infty} x_s[n]\, e^{-j\Omega n}$$

Now by comparing the CTFT impulse train and DTFT of the original sample sequence, we get the following equation.

$$W(\omega) = X_s(T_s\omega)$$

ω : CT frequency [rad/s]

Ω : DT frequency [rad/sample]

Step ② : Filling the gaps via convolution [Frequency - Domain]

In the time domain, we used $y(t) = h(t) * w(t)$

From Week 8, we know that Convolution in time ⟷ Multiplication in frequency

∴ Processing the signal through the filter in the frequency domain is just simple multiplication :

$$Y(\omega) = H(\omega)\,W(\omega) \xrightarrow{W(\omega) = X_s(T_s\omega)} Y(\omega) = H(\omega)\,X_s(T_s\omega)$$

Ideal Reconstruction in Frequency-Domain

So far we built the general frequency-domain reconstruction model:

$$Y(\omega) = H(\omega)W(\omega), \text{ where } W(\omega) = X_s(T_s\omega)$$

From our previous section, we know the bridge btwn DT and CT is $\Omega = \omega T_s$ and $W(\omega) = X_s(T_s\omega)$. Subtituiting into the DT sample formula:

$$X_s(\Omega) = \frac{1}{T_s}\sum_{\ell=-\infty}^{\infty} X\left(\frac{\Omega - 2\pi\ell}{T_s}\right)$$

$$\downarrow$$

$$\boxed{W(\omega) = \frac{1}{T_s}\sum_{\ell=-\infty}^{\infty} X(\omega - \ell\omega_s)} \quad \text{where} \quad \omega_s = \frac{2\pi}{T_s}$$

If we expand this sum, $W(\omega)$ is just an infinite string of scaled copies of our original spectrum centered at every multiple of ω_s

$$\cdots \frac{1}{T_s}X(\omega+\omega_s) + \boxed{\frac{1}{T_s}X(\omega)} + \frac{1}{T_s}X(\omega-\omega_s) \cdots$$

As long as we sampled fast enough ($\omega_s > 2W$), these copies do not overlap (alias). Sampling did not destroy the original spectrum.

Designing the ideal filter

To get our original $X(w)$ back, our filter $H(w)$ must do 2 jobs:

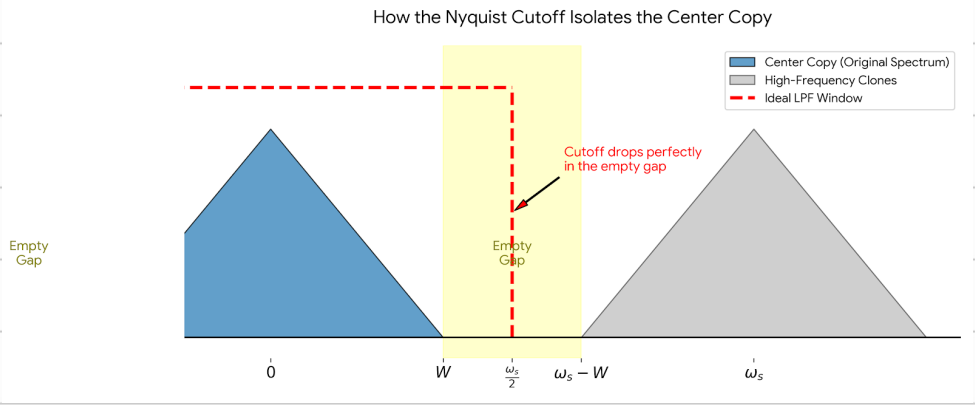
1) Keep the center copy & destroy all other copies
2) Rescale the height back to normal.

$$H_{ideal}(w) = \begin{cases} T_s, & |w| < \frac{\pi}{T_s} \\ 0, & \text{otherwise} \end{cases}$$

How this filter accomplishes the 2 jobs:

1) The cutoff ($w_c = \frac{\pi}{T_s} = \frac{w_s}{2}$): Since we followed the no aliasing rule, the Nyquist frequency drops perfectly in the middle of the empty gap. This places a window perfectly around the center copy while rejecting all high frequency clones at $\pm w_s$, $\pm 2w_s$, ... [Refer to illustration below]

2) The Gain of T_s: Because sampling each copy crushed the amplitude by a factor of $1/T_s$, the filter multiplies by T_s to perfectly cancel it out. $T_s \times \left(\frac{1}{T_s} X(w)\right) = X(w)$



$\therefore Y(w) = H_{ideal}(w)\, W(w) = X(w)$

$\Rightarrow y(t) = x(t)$

Essentially, if there is no aliasing the original spectral shape is perfectly intact inside the center copy. H_{ideal} isolates & rescales this copy back to the exact original spectrum, $X(w)$. Bcuz the spectrum is fully restored in the frequency domain, we have perfectly recovered the original CT signal $x(t)$ in the time domain. [Perfect reconstruction].

Ideal Reconstruction in the Time Domain

We just designed the perfect frequency-domain filter (Rectangular LPF).
Now we translate that exact same filter back into the time domain.

Recall Week 8 LPF, we already know the filwing:

Rectangular Freq. Response $\longleftrightarrow$ Sinc-shaped Impulse Response

$\therefore$ Since our $H_{ideal}(w)$ is a perfect rectangle, our time-domain reconstruction pulse $h_{ideal}(t)$ must be a sinc wave

$$h_{ideal}(t) = \text{sinc}\left(\frac{t}{T_s}\right) = \frac{\sin(\pi t/T_s)}{\pi t/T_s} \quad \text{where} \quad \text{sinc}(x) = \frac{\sin(\pi x)}{\pi x}$$

Recall the general reconstruction formula [Connect the dots formula]:

$$y(t) = \sum_n x_s[n]\, h(t - nT_s)$$

By choosing $h(t) = h_{ideal}(t)$:

Sinc Interpolation $$y(t) = x(t) = \sum_{n=-\infty}^{\infty} x_s[n]\, \text{sinc}\left(\frac{t - nT_s}{T_s}\right)$$

The Sinc wave is the "perfect" pulse bcuz it naturally obeys the Interpolation Condition we established earlier ($h(kT_s) = \delta[k]$).

- At its own centre ($t = 0$), $\text{sinc}(0) = 1$
- At every neighbouring sample spacing ($t = 1T_s, 2T_s, \dots$), the wave perfectly crosses the zero axis: $\text{sinc}(\pm 1) = \text{sinc}(\pm 2) = \dots = 0$

Table Summary

Frequency-Domain View	Time-Domain View
$H_{ideal}(w)$ is an ideal Rectangular LPF	$h_{ideal}(t)$ is a continuous Sinc Wave
Isolate original centre spectrum	Fill in empty time gaps [Sinc Interpolation]

The Complete Picture

Earlier, we established $\omega_s > 2W$ as the no-aliasing condition. Now that we understand reconstruction, we see the full chronological story of why the rule is the key to everything:

$\omega_s > 2W$

↓

Spectral copy remain separated

↓

An LPF can perfectly isolate the original center copy

↓

$y(t) = x(t)$ [Perfect Reconstruction]

The Big Takeaway is:

Sampling Theorem ⇒ Tells us WHEN reconstruction is possible

Sinc Interpolation ⇒ Tells us HOW to actually do it.

Practical Reconstruction

In Week 8 we established that a sinc impulse response is non-causal since it extends infinitely in both directions (it exist for $t < 0$).

Bcuz we can't see into the future, we have to compromise. The compromise being the trade off btwn reconstruction quality and delay.

Filter Type	How It Works	Causality	Effective Delay	Trade-off / Result
Zero-Order Hold (ZOH)	Uses only the current sample to hold a flat horizontal line.	100% Causal	$\approx \frac{T_s}{2}$	Very simple to build, but creates a rough, blocky staircase wave.
First-Order Hold (FOH)	Draws a straight sloped line from the current sample to the *next* sample.	Non-causal (naturally) Causal (if delayed)	T_s	Smoother curve. To make it causal, the system must wait T_s seconds for the next sample to arrive before drawing the line.
Ideal Sinc (Brick-wall LPF)	Uses perfectly scaled, overlapping continuous sinc pulses.	Strictly Non-Causal	Infinite (∞)	Mathematically perfect reconstruction, but physically impossible to build (requires knowing infinitely many future samples).

WEEK 10

Laplace Transform

Laplace Transform [Extension of CTFT]

The Continuous-Time Fourier Transform (CTFT) only works if a signal is absolutely integrable ($\int |x(t)|\,dt < \infty$). If a signal grows over time (e.g. $e^{t}u(t)$), the Fourier Integral blows up to infinity and fails.

To solve this, we multiply the signal by a **real exponential decay weight**, $e^{-\sigma t}$ before we transform it.

$$x_\sigma(t) = x(t)\,e^{-\sigma t}$$

$$\downarrow \text{CTFT}$$

$$X_\sigma(w) = \int_{-\infty}^{\infty} x_\sigma(t)\,e^{-jwt}\,dt$$

$$= \int_{-\infty}^{\infty} [x(t)e^{-\sigma t}]\,e^{-jwt}\,dt$$

$$X_\sigma(w) = \int_{-\infty}^{\infty} x(t)\,e^{-(\sigma+jw)t}\,dt$$

$$\downarrow \hat{X}(w) = X_\sigma(w)$$

Bilateral Laplace Transform $\hat{X}(s) = \int_{-\infty}^{\infty} x(t)e^{-st}\,dt$ where $s = \sigma + jw$

- σ: Real part [Decay]
- jw: Imaginary [Fourier freq.]

Essentially, the Laplace Transform is just the CTFT equipped with an exponential safety net ($e^{-\sigma t}$). It allow us to analyze growing signals that the standard Fourier Transform can't handle.

Region of Convergence (ROC)

Adding an exponential weighing factor doesn't guarantee convergence for every s. For instance, take the following example:

$$x(t) = e^{t}\,u(t)$$

↓ Laplace Transform

$$\hat{X}(s) = \int_0^{\infty} e^{t} e^{-st}\,dt$$

$$= \int_0^{\infty} e^{-(s-1)t}\,dt \quad \text{— ①}$$

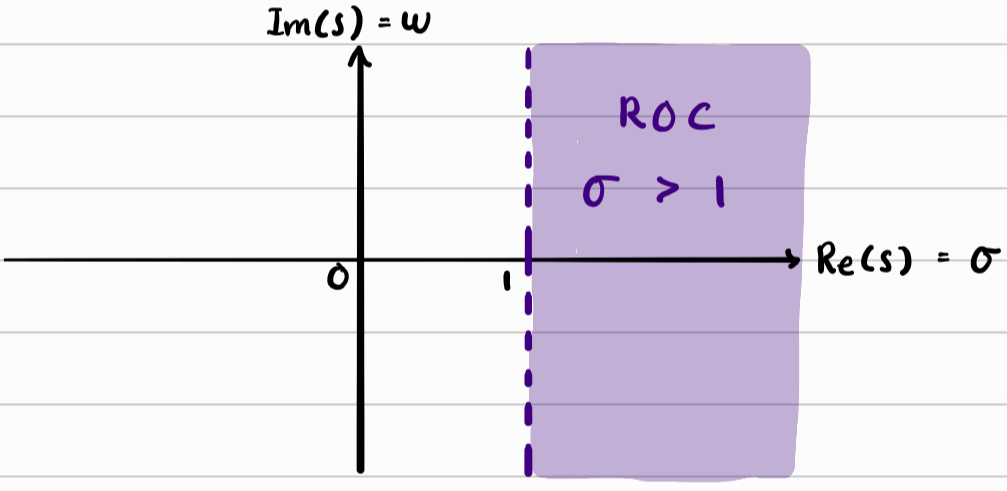
Notice that this integral only converges when

$$Re(s-1) > 0 \implies Re(s) > 1$$

Only under this condition, Eqn ① can be safetly evaluated

Region of Convergence (ROC)

$$\hat{X}(s) = \frac{1}{s-1}, \quad Re(s) > 1$$



∴ The valid territory of s values is called ROC:

$$RoC(x) = \{ s \in C : \int_{-\infty}^{\infty} |x(t)e^{-st}|\,dt < \infty \}$$

weighted signal absolutely integrable

Since e^{-jwt} is just a pure rotation with a constant magnitude of 1

$$\implies |e^{-st}| = |e^{-\sigma t} e^{-jwt}| = e^{-\sigma t}$$

∴ Whether the integral blows up or converges depends only on the real decay part, σ.

Connecting Laplace Transform back to CTFT

Recall : $s = \sigma + j\omega$, if we set $\sigma = 0$

$\Longrightarrow$ $s = j\omega$, so the Laplace Transform becomes

$$\hat{X}(s) = \hat{X}(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} \, dt$$

$$\boxed{\hat{X}(j\omega) = X(\omega)}$$, if $\sigma = 0$ is inside ROC .

This is quite logical , since Laplace Transform is literally just CTFT with the $e^{-\sigma t}$ safety net . When we set $\sigma = 0$, we remove the safety net , thus becoming back to CTFT .

Take Note :

This relationship is only valid if the imaginary axis ($s = j\omega$) lies inside the Region of Convergence (RoC)

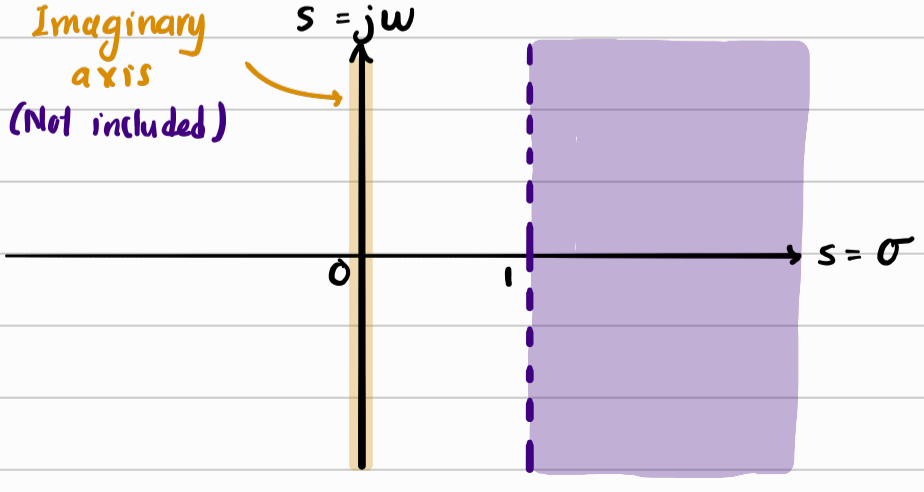
For example , for $x(t) = e^{t} u(t)$, we found

$$\hat{X}(s) = \frac{1}{s-1} \quad , \quad \text{RoC}(s) > 1 \quad (\sigma > 1)$$

Notice that the imaginary axis ($s = j\omega$) is outside the RoC , therefore saying

$$\hat{X}(j\omega) = \frac{1}{j\omega - 1} = X(\omega)$$

is invalid , beuz the CTFT for $e^{t} u(t)$ blows up and doesn't converge in the first place



Laplace Transform of Complex Exponentials

Now we understand that a valid, complete Laplace transform must be paired up with its RoC. To see why this pairing is absolutely crucial, let's look at the complex exponential.

Right-sided Complex Exponential

Sample Signal : $x_R(t) = e^{-\alpha t} u(t)$

↓ Laplace Transform

$$\hat{X}_R(s) = \int_0^\infty e^{-\alpha t} e^{-st}\, dt$$

$$= \int_0^\infty e^{-(s+\alpha)t}\, dt$$

For this to converge as $t \to \infty$

$\Rightarrow \text{Re}(s+\alpha) > 0$

$\Rightarrow \text{Re}(s) > -\text{Re}(\alpha)$

$\therefore$ Evaluating under this condition

$\Rightarrow \hat{X}_R(s) = \frac{1}{s+\alpha}$

RoC : $\text{Re}(s) > -\text{Re}(\alpha)$

Left Sided Complex Exponential

Sample Signal : $x_L(t) = -e^{-\alpha t} u(-t)$

↓ Laplace Transform

$$\hat{X}_R(s) = -\int_{-\infty}^0 e^{-\alpha t} e^{-st}\, dt$$

$$= -\int_{-\infty}^0 e^{-(s+\alpha)t}\, dt$$

For this to converge as $t \to -\infty$

$\Rightarrow \text{Re}(s+\alpha) < 0$

$\Rightarrow \text{Re}(s) < -\text{Re}(\alpha)$

$\therefore$ Evaluating under this condition

$\Rightarrow \hat{X}_R(s) = \frac{1}{s+\alpha}$

RoC : $\text{Re}(s) < -\text{Re}(\alpha)$

Notice that the exact same algebraic Laplace expression can represent completely different time domain signals. The RoC is the only thing that tells them apart.

Rational Laplace Transforms

The example above produced expressions like

$$\hat{X}(s) = \frac{1}{s+\alpha}$$

These are example of **rational function** — ratio of polynomial in s

More generally, many Laplace Transforms have the form

$$\hat{X}(s) = \frac{B(s)}{A(s)} = \frac{b_0 + b_1 s + b_2 s^2 + \dots + b_m s^m}{a_0 + a_1 s + a_2 s^2 + \dots + a_n s^n}$$

where $B(s)$ and $A(s)$ are polynomials in s.

Assuming any common factor have been cancelled, we can identify 2 important features:

- Zeros : Roots of $B(s) \Longrightarrow \hat{X}(s) = 0$
- Poles : Roots of $A(s) \Longrightarrow \hat{X}(s) =$ undefined

Since a pole is a location where the algebraic Laplace expression becomes undefined, so the Laplace integral cannot converge there.

$\Longrightarrow$ RoC cannot contain any poles

This also explains why poles are particularly useful when determining RoC, they help establish the boundaries of the RoC.

Region of Convergence and Signal relationship

We already know that the Laplace Transform only converges for certain values of $s = \sigma + j\omega$, and that convergence depends on the real part σ.

Additionally, the region where a signal exist in time also tells us which exponential weighting can make its Laplace Transform converge.

① Bounded Finite-Duration Signals

A finite-duration signal is zero outside some finite interval $[a, b]$.

If the signal is absolutely integrable, then multiplying it by $e^{-\sigma t}$ won't introduce infinite growth over that finite interval for any finite σ.

Therefore,

Complex Number

$$RoC(x) = \mathbb{C}$$

meaning the entire complex s-plane.

② Right-Sided Signal

A right-sided signal is zero before some time a:

$$x(t) = 0, \quad t < a$$

Since the signal may extend infinitely toward $+\infty$, we need sufficient exponential decay in that direction

Therefore

real value convergence boundary

$$Re(s) > p$$

For rational Laplace Transforms:

RoC lies to the RIGHT of rightmost pole

③ Left sided Signal

A left-sided signal is zero after some time a:

$$x(t) = 0 \quad , \quad t > a$$

Now the signal may extend infinately toward $-\infty$, so we need the exponential weighting to control the negative-time tail

Therefore

$$Re(s) < p$$

real value convergence boundary

For rational Laplace Transforms:

RoC lies to the LEFT of the leftmost pole

④ Two-sided signals

A two sided signal extends infinately in both time directions.

$$x(t) = x_R(t) + x_L(t)$$

For the complete Laplace Transform to exist, both sides must converge simultaneously.

So the RoC is the intersection of their two convergence region:

$$RoC(x) = RoC(x_R) \cap RoC(x_L)$$

- Overlap → $p_1 < Re(s) < p_2$ (Vertical strip)
- No Overlap → $RoC(x) = \varnothing$ (empty)

Properties of the Laplace Transform

Now that we understand Laplace Transform and their RoCs, we can reuse many of the Fourier-Transform properties from Week 8.

The main difference is that we must now also consider how the operation affects RoC.

① Linearity

If $y(t) = a x_1(t) + \beta x_2(t)$

$$\hat{Y}(s) = a\hat{X}_1(s) + \beta\hat{X}_2(s), \quad \text{RoC}(y) \supseteq \text{RoC}(x_1) \cap \text{RoC}(x_2)$$

superset

Why $\supseteq$ instead of $=$

The signal may cancel each other, allowing their sum to converge over a larger region

② Time Scaling

If $y(t) = x(at)$, $a > 0$

$$\hat{Y}(s) = \frac{1}{a}\hat{X}\left(\frac{s}{a}\right), \quad \text{RoC}(y) = \left\{s : \frac{s}{a} \in \text{RoC}(x)\right\}$$

Read as

"The RoC of y equals the set of all s such that $\frac{s}{a}$ is an element of the RoC of x"

③ Modulation

Recall from CTFT that multiplying a signal by a complex exponential shift its frequency representation. Laplace transform extends this idea:

$y(t) = e^{s_0 t}\, x(t)$, where s_0 is a complex constant.

Taking the Laplace Transform,

$$\hat{Y}(s) = \int_{-\infty}^{\infty} x(t)\, e^{s_0 t}\, e^{-st}\, dt$$
$$= \int_{-\infty}^{\infty} x(t)\, e^{-(s-s_0)t}\, dt$$

$$\hat{Y}(s) = \hat{X}(s - s_0)$$

$$\text{RoC}(y) = \{ s : s - s_0 \in \text{RoC}(x) \}$$

Essentially, multiplication by $e^{s_0 t}$ shifts the transform representation in the complex s-plane.

Unlike ordinary Fourier modulation, s_0 can have a real part, so the shift can also change the convergence boundary.

④ Convolution

We already know that convolution in time becomes multiplication in the frequency domain. The same mechanism holds for Laplace Transform.

$$y(t) = x_1(t) * x_2(t)$$

$$\hat{Y}(s) = \hat{X}_1(s)\, \hat{X}_2(s)$$

$$\text{RoC}(y) \supseteq \text{RoC}(x_1) \cap \text{RoC}(x_2)$$

superset

Again cuz cancellation can occur causing RoC(y) to be larger than the intersection

⑥ Multiplication by t

Suppose $y(t) = t\,x(t)$,

Since $\hat{X}(s) = \int_{-\infty}^{\infty} x(t)\, e^{-st}\, dt$

$\downarrow d/ds$

$$\frac{d}{ds}\hat{X}(s) = \int_{-\infty}^{\infty} x(t)\,(-t)\, e^{-st}\, dt$$

$$-\frac{d}{ds}\hat{X}(s) = \int_{-\infty}^{\infty} [t\,x(t)]\, e^{-st}\, dt$$

$\therefore$ $\boxed{\hat{Y}(s) = -\frac{d}{ds}\hat{X}(s)}$ $\qquad$ $\boxed{RoC(y) = RoC(x)}$

WEEK 11

Laplace Transform and CT LTI system

Transfer Function of an LTI system

Recall from Week 5-6 that we could describe an LTI system using its frequency response. We deliberately applied a complex exponential to the system:

$$x(t) = e^{jwt}$$

$$\downarrow \text{BIBO stable LTI system}$$

$$y(t) = H(w)\, e^{jwt}$$

The important idea was that a complex exponential passes through an LTI system without changing its frequency. The system only introduce a complex scaling factor $H(w)$.

But from Week 10, we learned that Laplace Transform extends the Fourier-Transform viewpoint by introducing

$$s = \sigma + jw$$

So instead of using the purely oscillating exponential e^{jwt}, we apply the more general complex exponential

$$x(t) = e^{st} = e^{\sigma t} e^{jwt}$$

Fortunately, the same LTI mechanism still applies

$$e^{st} \xrightarrow{\text{LTI system}} \hat{H}(s)\, e^{st}$$

$\hat{H}(s)$: Transfer Function

Transfer Function $\hat{H}(s)$

We already know that every LTI system is characterised by its impulse response $h(t)$:

$$y(t) = h(t) * x(t)$$

Using convolution,

$$y(t) = \int_{-\infty}^{\infty} h(\tau)\, x(t-\tau)\, d\tau$$

Let test input, $x(t) = e^{st}$,

$$y(t) = \int_{-\infty}^{\infty} h(\tau)\, x(t-\tau)\, d\tau$$

$$= \int_{-\infty}^{\infty} h(\tau)\, e^{s(t-\tau)}\, d\tau$$

$$y(t) = \underbrace{\left[\int_{-\infty}^{\infty} h(\tau)\, e^{-s\tau}\, d\tau\right]}_{\hat{H}(s)} e^{st}$$

Therefore,

$$\hat{H}(s) = \int_{-\infty}^{\infty} h(t)\, e^{-st}\, dt \quad \text{for } s \text{ in the RoC of } h$$

Relationship to frequency response

Recall that frequency response used e^{jwt}, whereas transfer function uses e^{st}

By choosing $\sigma = 0$, we obtain $s = jw$. Therefore

$$H(w) = \hat{H}(jw) \quad \text{if the } \underbrace{\text{imaginary axis}}_{\sigma = 0} \text{ is within RoC}$$

Input - Output Relationship in the Laplace Domain

The response to e^{st} explains what $\hat{H}(s)$ represents. But we don't want to restrict ourselves to exponential inputs.

Recall the original time domain relationship:

$$y(t) = h(t) * x(t)$$

from the convolution property of Laplace Transforms,

$$\mathcal{L}\{h * x\} = \mathcal{L}\{h\}\mathcal{L}\{x\}$$

Therefore,

$$\hat{Y}(s) = \hat{H}(s)\,\hat{X}(s) \quad \text{OR} \quad \hat{H}(s) = \hat{Y}(s)/\hat{X}(s)$$

As expected, these holds where the relevant transforms are valid. We must still account for their RoCs.

> Concept Check:
>
> This shouldn't be new, it is literally just the same as what we learned in Week 8
>
> $$Y(\omega) = H(\omega)\,X(\omega)$$
>
> But now its just Laplace Transform
>
> $$\hat{Y}(s) = \hat{H}(s)\,\hat{X}(s)$$

Transfer Function of Familiar System

The mechanism to get the transfer function stays the same

↳ Apply e^{st} as the test input $[x(t) = e^{st}]$

① Delay System

For a delay of τ : $y(t) = x(t-\tau)$

$$\downarrow x(t) = e^{st}$$

$$y(t) = e^{s(t-\tau)} = e^{-s\tau} e^{st}$$

$$\hat{H}(s) = e^{-s\tau}$$

RoC : Entire Complex plane

② Differentiator System

For a differentiator : $y(t) = \frac{d}{dt} x(t)$

$$\downarrow x(t) = e^{st}$$

$$y(t) = s\, e^{st}$$

$$\hat{H}(s) = s$$

③ Integrator System

For causal integrator : $y(t) = \int_{-\infty}^{t} x(\tau)\, d\tau$

$$\hat{H}(s) = \frac{1}{s}$$

RoC : $\mathrm{Re}(s) > 0$

④ System described by differential equations

Suppose an initial-rest CT LTI system is governed by $\sum_{k=0}^{N} a_k \frac{d^k y}{dt^k} = \sum_{m=0}^{M} b_m \frac{d^m x}{dt^m}$

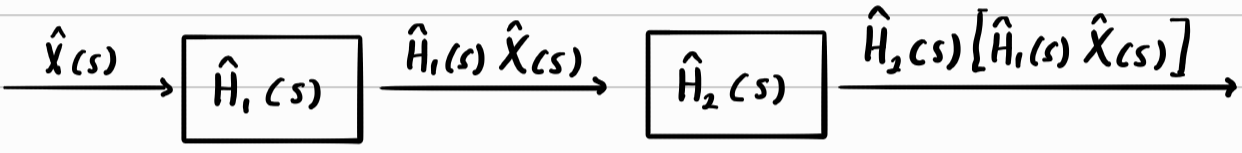
$$\hat{H}(s) = \frac{\sum_{m=0}^{M} b_m s^m}{\sum_{k=0}^{N} a_k s^k}$$

Transfer Functions of Interconnected LTI system

Back in Week 2, we learned that simple building block systems can be combined into larger systems.
Week 3 then showed that interconnected LTI system can be described through combinations of their impulse responses.

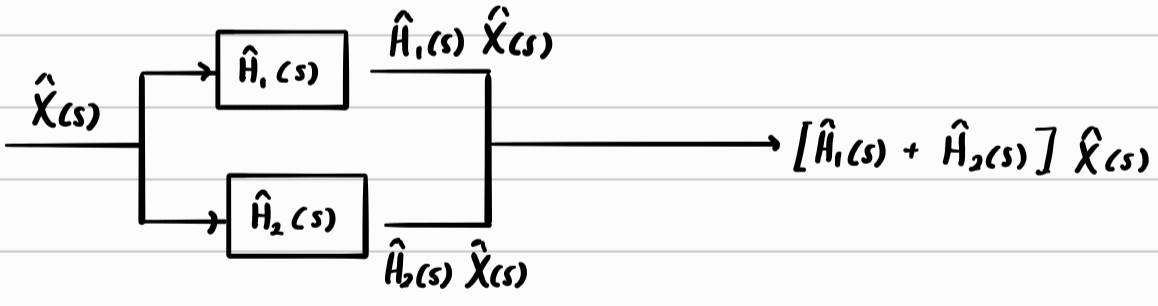
Now we do the same thing using transfer functions.

① Series / Cascade interconnection



$$\hat{H}_{series}(s) = [\hat{H}_1(s)][\hat{H}_2(s)]$$

② Parallel interconnection



$$\hat{H}_{parallel}(s) = \hat{H}_1(s) + \hat{H}_2(s)$$

③ Feedback

Feedback introduces one extra idea : Part of the output is returned to influence the input of the forward system.

RoC, Causality and BIBO stability

Previously the RoC helped us distinguish btwn different signals have the same algebraic Laplace Transform.

Now the signal being transformed is the impulse response, $h(t)$.

Since $h(t)$ characterises the entire LTI system, its RoC can tell us about the system's properties.

① Causality

Recall from Week 3 that an LTI system is causal if and only if its impulse response is zero before the impulse occurs:

$$h(t) = 0, \quad t < 0$$

So a causal system's impulse response is right sided. From Week 10 we know that the RoC of a typical right-sided signal extends to the right of its poles.

$$\text{RoC}: \operatorname{Re}(s) > \operatorname{Re}(p_R)$$

p_R: rightmost pole

② BIBO stability

For an ordinary CT LTI system, the familiar impulse-response is:

$$\int_{-\infty}^{\infty} |h(t)|\, dt < \infty$$

But this is exactly the absolute-integrability condition that allows us to evaluate the ordinary Fourier transform of $h(t)$.

And from Week 10, we know that

$$H(\omega) = \hat{H}(j\omega), \text{ if } \sigma = 0 \text{ is within RoC}$$

Therefore,

BIBO stable $\longleftrightarrow$ RoC of $\hat{H}(s)$ contains imaginary axis

Combining causality & stability

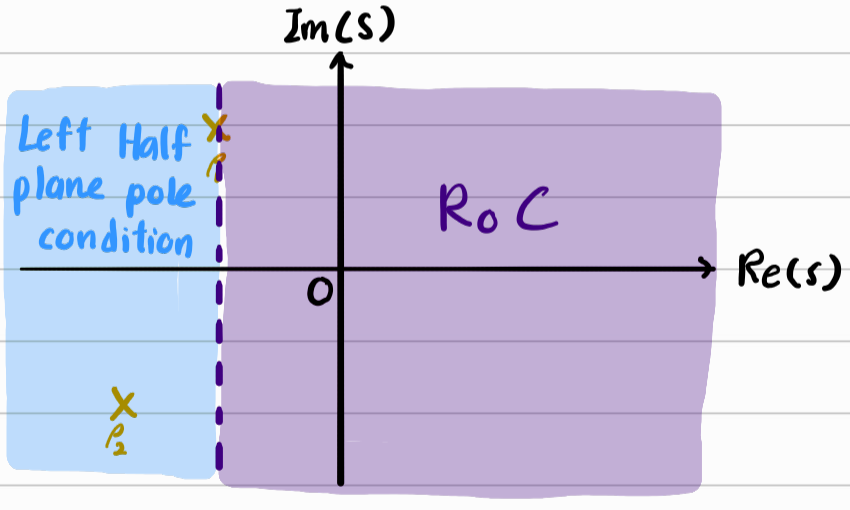
Now we combine the 2 conditions:

↳ Causal : RoC extend to the right of rightmost pole

↳ BIBO : RoC must include imaginary axis ($Re(s) = 0$)

Therefore, the rightmost pole must be strictly to the left of zero.

Causal rational CT LTI system is BIBO stable $\longleftrightarrow$ $Re(p_i) < 0$ for every pole p_i



Essentially, for a causal, BIBO stable system, placing all poles strictly to the left of the imaginary axis gives a RoC that includes the imaginary axis.

Left Half Plane pole condition

Inverse Laplace Transform

So far, Laplace Transform has made it easy to calculate the output representation: $\hat{Y}(s) = \hat{H}(s)\hat{X}(s)$

But the physical output is still $y(t)$, not $\hat{Y}(s)$. This is where inverse Laplace Transform comes in.

The general inverse Laplace formula involves integration, but ECE 2111 focuses on decomposing the signal into simpler terms and using the corresponding tables of inverse Laplace transform to compute it.

Laplace transform tables

Time domain signal $x(t)$	Laplace transform $\hat{X}(s)$	Region of convergence RoC(x)
$\delta(t-\tau)$	$e^{-s\tau}$	$\mathbb{C}$
$u(t)$	$\frac{1}{s}$	$\{s \in \mathbb{C} : \mathrm{Re}(s) > 0\}$
$\cos(\omega_0 t)u(t)$	$\frac{s}{s^2+\omega_0^2}$	$\{s \in \mathbb{C} : \mathrm{Re}(s) > 0\}$
$\sin(\omega_0 t)u(t)$	$\frac{\omega_0}{s^2+\omega_0^2}$	$\{s \in \mathbb{C} : \mathrm{Re}(s) > 0\}$
$e^{-\alpha t}u(t)$	$\frac{1}{s+\alpha}$	$\{s \in \mathbb{C} : \mathrm{Re}(s) > -\mathrm{Re}(\alpha)\}$
$-e^{-\alpha t}u(-t)$	$\frac{1}{s+\alpha}$	$\{s \in \mathbb{C} : \mathrm{Re}(s) < -\mathrm{Re}(\alpha)\}$
$\frac{t^{n-1}}{(n-1)!}e^{-\alpha t}u(t)$	$\frac{1}{(s+\alpha)^n}$	$\{s \in \mathbb{C} : \mathrm{Re}(s) > -\mathrm{Re}(\alpha)\}$
$-\frac{t^{n-1}}{(n-1)!}e^{-\alpha t}u(-t)$	$\frac{1}{(s+\alpha)^n}$	$\{s \in \mathbb{C} : \mathrm{Re}(s) < -\mathrm{Re}(\alpha)\}$

Table 1: Table of Laplace transform pairs. In the table, τ is a real number, ω_0 is a real number, α is a complex number, and n is a positive integer.

Example 1

Given $\hat{H}(s) = \frac{s-1}{(s+1)(s-2)}$, $\mathrm{Re}(s) > 2$ — right sided

Using Partial Fraction

$$\frac{s-1}{(s+1)(s-2)} = \frac{A_1}{s+1} + \frac{A_2}{s-2}$$

$$s - 1 = A_1(s-2) + A_2(s+1)$$

When $s = -1 \Rightarrow A_1 = \frac{s-1}{s-2} = \frac{-1-1}{-1-2} = \frac{2}{3}$

$s = 2 \Rightarrow A_2 = \frac{s-1}{s+1} = \frac{2-1}{2+1} = \frac{1}{3}$

$$\therefore \hat{H}(s) = \frac{2}{3}\left[\frac{1}{s+1}\right] + \frac{1}{3}\left[\frac{1}{s-2}\right]$$

Since the RoC is to the right of both poles

$$\Rightarrow h(t) = \frac{2}{3}e^{-t}u(t) + \frac{1}{3}e^{2t}u(t) \ \#$$

Example 2

Given $\hat{X}_s = \frac{s^3 + s^2 + s + 1}{s(s+2)^2}$, $-2 < Re(s) < 0$

Polynomial division and partial fraction give

$\Rightarrow \hat{X}(s) = 1 + \frac{1}{4}\left(\frac{1}{s}\right) + \frac{13}{4}\left(\frac{1}{s+2}\right) + \frac{5}{2}\left(\frac{1}{(s+2)^2}\right)$

$-2 < Re(s) < 0$

Now applying RoC :

$1 \longrightarrow \delta(t)$

$\frac{1}{s} \longrightarrow$ pole at $s=0$ $\longrightarrow$ left sided inverse $\longrightarrow u(-t)$

$\frac{1}{(s+2)} \longrightarrow$ pole at $s=-2$ $\longrightarrow$ right sided inverse $\longrightarrow e^{-2t}u(t)$

$\frac{1}{(s+2)^2} \longrightarrow$ pole at $s=-2$ $\longrightarrow$ right sided inverse $\longrightarrow te^{-2t}u(t)$

Therefore,

$$x(t) = \delta(t) - \frac{1}{4}u(-t) - \frac{13}{4}e^{-2t}u(t) + \frac{5}{2}te^{-2t}u(t) \quad \#$$

Essentially the general method to solve these question are :

Decompose the expression

↓

Invert individual terms

↓

Use RoC to choose correct time-domain form

Computing the Responses of General LTI system

From Week 3, the time-domain method was

$$y(t) = h(t) * x(t)$$

Now the Laplace-domain method is

$$\hat{Y}(s) = \hat{H}(s)\,\hat{X}(s)$$

These 2 approaches produce the same output, whenever the operations are valid.

The practical sequence is therefore:

Find $\hat{H}(s)$

↓

Find $\hat{X}(s)$

↓

$\hat{Y}(s) = \hat{H}(s)\,\hat{X}(s)$

↓

Use RoC and inverse Laplace Transform to recover $y(t)$

WEEK 12

Z transform
&
DT LTI System

Z-Transform Introduction

Recall from Week 8 that the DTFT is

$$X(w) = \sum_{n=-\infty}^{\infty} x[n]\, e^{-jwn}$$

where w is the frequency being examined, while n scans through all the samples of the DT signal.

We previously learned that one sufficient condition for this infinite sum to converge is **absolute summability**:

$$\sum_{n=-\infty}^{\infty} |x[n]| < \infty$$

But what if the **signal grows instead of decays**? This is exactly the limitation we encountered with the **CTFT** in Week 10. Back then, we solved the problem by introducing **Laplace Transform**. We will now use the same approach for DT

Z - Transform

Recall how **Laplace Transform** was developed. We multiplied the original **CT signal** by a **real exponential weighting factor**, $e^{-\sigma t}$. If the weighting makes the signal **absolutely integrable**, we can take the **Fourier Transform of the weighted signal.**

Now we apply the same idea to **DT signal**. Instead of $e^{-\sigma t}$, we introduce the weighting factor r^{-n}, where $r > 0$

$$x_r[n] = x[n]\, r^{-n}$$

$$\downarrow \text{DTFT}$$

$$X_r(\omega) = \sum_{n=-\infty}^{\infty} x_r[n]\, e^{-j\omega n}$$

$$= \sum_{n=-\infty}^{\infty} x[n] r^{-n} e^{-j\omega n}$$

$$X_r(\omega) = \sum_{n=-\infty}^{\infty} x[n] \cdot (re^{j\omega})^{-n}$$

$$\downarrow \hat{X}(z) = X_r(\omega)$$

Bilateral Z - transform $\hat{X}(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$ where $z = r\, e^{j\omega}$

r: magnitude; $e^{j\omega}$: Angle

Concept Check:

	Laplace transform (CT)	Z-transform (DT)
Complex variable	$s = \sigma + j\omega$	$z = re^{j\omega}$
Weighting	$e^{-\sigma t}$	r^{-n}
Frequency analysis	$e^{-j\omega t}$	$e^{-j\omega n}$
Transform	$\int x(t)e^{-st}dt$	$\sum x[n]z^{-n}$

The underlying mechanism hasn't changed. We are still analysing frequency content, but we first modify the signal with an exponential weighting to make the transform converge.

Region of Convergence (RoC)

Just like Laplace Transform, the Z-transform does not necesarilly converge for every complex value of z.

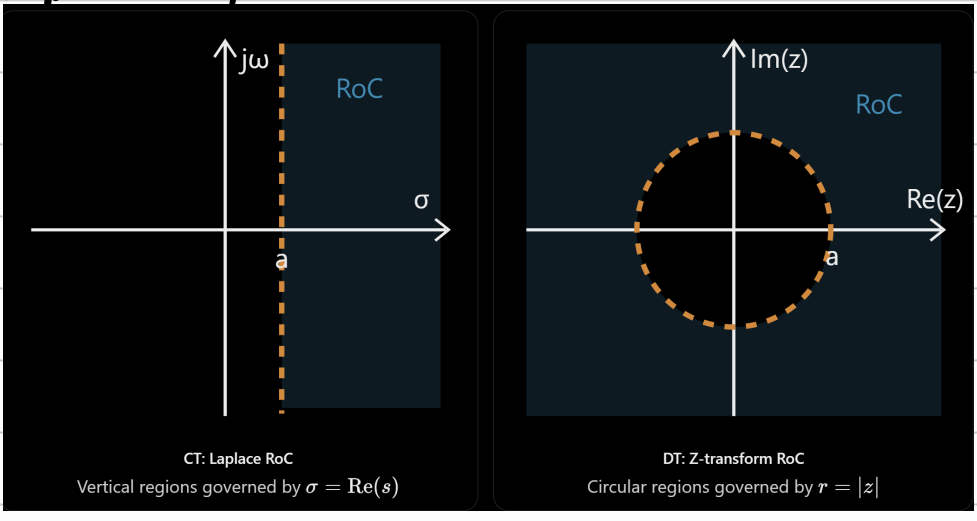
The Region of Convergence is the set of all z-values for which the weighted signal is absolutely summable:

$$RoC(x) = \left\{ z \in \mathbb{C} : \sum_{n=-\infty}^{\infty} |x[n]\, z^{-n}| < \infty \right\}$$

The important thing to notice is that ($|e^{-j\omega n}| = 1$)

$$|z^{-n}| = |r^{-n}|\,|e^{-j\omega n}| = r^{-n}$$

Therefore, absolute convergence depends on the radius ($r = |z|$), not angle (ω)
This explains why Z-transform RoCs are circular regions in the complex z-plane



Example for Right-Sided Cases

Relationship btwn Z-transform and DTFT

Recall: $z = re^{j\omega}$, if we set $r = 1$

$\Longrightarrow$ $z = e^{j\omega}$, so the Z-transform becomes

$$\hat{X}(z) = \hat{X}(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n](e^{j\omega})^{-n}$$

$$= \sum_{n=-\infty}^{\infty} x[n]\, e^{-j\omega n}$$

$$\hat{X}(e^{j\omega}) = X(\omega)$$

if unit circle ($z = e^{j\omega}$) lies inside RoC

Z-transform of Complex Exponential

The following examples demonstrate the same important idea in Week 10:

The same algebraic transform expression can represent different time-domain signals, depending on the RoC.

Right Sided Complex Exponential

Sample Signal : $x_R[n] = a^n u[n]$

$\downarrow$ Z-transform

$$\hat{X}_R(z) = \sum_{n=-\infty}^{\infty} a^n u[n] z^{-n}$$
$$= \sum_{n=0}^{\infty} (az^{-1})^n$$

For this to converge as $n \to \infty$

$\Rightarrow |az^{-1}| < 1$

$\Rightarrow |z| > |a|$

$\therefore$ Using Geometric Series Formula:

$$\Rightarrow \hat{X}_R(z) = \frac{1}{1 - az^{-1}}$$
$$= \frac{z}{z - a}$$

$RoC_R : |z| > |a|$

Left Sided Complex Exponential

Sample Signal : $x_L[n] = -a^n u[-n-1]$

$\downarrow$ Z-transform

$$\hat{X}_L(z) = -\sum_{n=-\infty}^{-1} a^n z^{-n}$$

To apply geometric series formula, substitute $m = -n$

$$\Rightarrow \hat{X}_L(z) = -\sum_{m=1}^{\infty} \left(\frac{z}{a}\right)^m$$

For this to converge as $n \to \infty$

$\Rightarrow \left|\frac{z}{a}\right| < 1$

$\Rightarrow |z| < |a|$

$\therefore$ Using Geometric Series Formula:

$$\Rightarrow \hat{X}_L(z) = -\frac{z/a}{1 - z/a}$$
$$= \frac{z}{z - a}$$

$RoC_L : |z| < |a|$

Notice that the exact same algebraic z-transform expression can represent completely different time domain signals. The RoC is the only thing that tells them apart.

General RoCs shapes

We've seen that RoC depends on the direction in which a signal extends. They can be generalised into the following cases.

Signal Type	CT Laplace RoC	DT z-transform RoC
Right-Sided	Right Half-plane	Exterior of circle
Left-Sided	Left Half-plane	Interior of circle
Two-sided	Vertical Strip	Annulus (Donut)

For a typical rational z-transform:

Right-sided	$\|z\| > r_1$
Left-sided	$\|z\| < r_2$
Two-sided	$r_1 < \|z\| < r_2$

Rational z-transform and poles

Recall from Week 10, that a Laplace transform is called rational when it can be expressed as a ratio of polynomials. Same definition here.

A rational z-transform has the form

$$\hat{X}(z) = \frac{B(z)}{A(z)} = \frac{b_0 + b_1 z^{-1} + \dots + b_M z^{-M}}{a_0 + a_1 z^{-1} + \dots + a_N z^{-N}}$$

Assuming any common factor have been cancelled, we can identify 2 important features:

- Zeros: Roots of $B(z) \Longrightarrow \hat{X}(z) = 0$
- Poles: Roots of $A(z) \Longrightarrow \hat{X}(z) =$ undefined

Similiar to Laplace domain, the RoC cannot contain any poles.
For example if the poles have magnitude 0.5 and 2,
the possible regions include

$|z| < 0.5$, $0.5 < |z| < 2$, $|z| > 2$

The actual region depends on the time domain signal sideness.

Z-transform properties

The lecture slide introduces 4 main Z-transform properties.
These aren't completly new rules, they are the DT equivalent of properties we already learned for the Laplace Transform.

Property	Time-domain operation	Z-domain result	RoC
Linearity	$y[n] = \alpha x_1[n] + \beta x_2[n]$	$\hat{Y}(z) = \alpha \hat{X}_1(z) + \beta \hat{X}_2(z)$	$RoC(y) \supseteq RoC(x_1) \cap RoC(x_2)$
Modulation	$y[n] = z_0^n x[n]$	$\hat{Y}(z) = \hat{X}(z/z_0)$	$RoC(y) = \{z : \frac{z}{z_0} \in RoC(x)\}$
Convolution	$y[n] = x_1[n] * x_2[n]$	$\hat{Y}(z) = \hat{X}_1(z)\hat{X}_2(z)$	$RoC(y) \supset RoC(x_1) \cap RoC(x_2)$
Conjugation	$y[n] = x^*[n]$	$\hat{Y}(z) = [\hat{X}(z^*)]^*$	Unchanged

Tranfer Function of DT LTI systems

Recall from Week 11, for a CT LTI system, we applied a test input

$$x(t) = e^{st}$$

and found that the output was

$$y(t) = \hat{H}(s)\, e^{st} \quad \text{, where} \quad \hat{H}(s) = \int_{-\infty}^{\infty} h(t)\, e^{-st}\, dt$$

In other words, the transfer function is the Laplace Transform of the impulse response.

Now we do the same thing for DT.

Recall that LTI system operates by convolution:

$$y[n] = \sum_{k=-\infty}^{\infty} h[k]\, x[n-k]$$

Let $x[n] = z^n$ be the test input

$$y[n] = \sum_{k=-\infty}^{\infty} h[k]\, z^{n-k}$$

$$= \underbrace{\left[\sum_{k=-\infty}^{\infty} h[k] z^{-k}\right]}_{\hat{H}(z)} z^n$$

Therefore,

$$\hat{H}(z) = \sum_{k=-\infty}^{\infty} h[k] z^{-k} \quad \text{for } z \text{ in the RoC of } h$$

Concept Check

	Continuous time	Discrete time
Test input	e^{st}	z^n
Output	$\hat{H}(s)e^{st}$	$\hat{H}(z)z^n$
Transfer function	Laplace transform of $h(t)$	Z-transform of $h[n]$
Frequency response	$H(\omega) = \hat{H}(j\omega)$	$H(\omega) = \hat{H}(e^{j\omega})$

The frequency response relationship apply when the relevant imaginary axis or unit circle is contained in the RoC.

Computing outputs in the $\mathcal{Z}$-domain

We don't want to restrict ourselves to the simple test input z^n.
We want to find the output for an arbitrary input signal.

Recall the general DT convolution relationship:

$$y[n] = h[n] * x[n]$$

From the $\mathcal{Z}$-transform convolution property:

$$\hat{Y}(z) = \hat{H}(z)\,\hat{X}(z)$$

We can apply the inverse $\mathcal{Z}$-transform, using the appropriate RoC, to recover $y[n]$. This is the exact same workflow as Week 11.

Find Input Transform
↓
Multiply by transfer function
↓
Inverse Transform using RoC
↓
Output

Z-transform tables

If x is a discrete-time signal then

$$\hat{X}(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$$

Time domain signal $x[n] =$	Z-transform $\hat{X}(z) =$	Region of convergence RoC(x)
$\delta[n]$	1	$\mathbb{C}$
$u[n]$	$\frac{1}{1-z^{-1}}$	$\{z \in \mathbb{C} \,:\, \lvert z\rvert > 1\}$
$nu[n]$	$\frac{z^{-1}}{(1-z^{-1})^2}$	$\{z \in \mathbb{C} \,:\, \lvert z\rvert > 1\}$
$a^n u[n]$	$\frac{1}{1-az^{-1}}$	$\{z \in \mathbb{C} \,:\, \lvert z\rvert > \lvert a\rvert\}$
$a^n u[-n]$	$\frac{1}{1-a^{-1}z}$	$\{z \in \mathbb{C} \,:\, \lvert z\rvert < \lvert a\rvert\}$
$na^n u[n]$	$\frac{az^{-1}}{(1-az^{-1})^2}$	$\{z \in \mathbb{C} \,:\, \lvert z\rvert > \lvert a\rvert\}$
$\delta[n-m]$	z^{-m}	$\lvert z\rvert > 0$
$a^n \cos[\omega_0 n]u[n]$	$\frac{1-az^{-1}\cos(\omega_0)}{1-2az^{-1}\cos(\omega_0)+a^2z^{-2}}$	$\lvert z\rvert > \lvert a\rvert$
$a^n \sin[\omega_0 n]u[n]$	$\frac{az^{-1}\sin(\omega_0)}{1-2az^{-1}\cos(\omega_0)+a^2z^{-2}}$	$\lvert z\rvert > \lvert a\rvert$

Table 1: Table of Z-transform pairs. In the table, the parameter a is a complex number, and the parameter ω_0 is a real number.

Time domain $y[n] =$	Z-transform $\hat{Y}(z) =$	Region of convergence RoC(y)	Name
$\alpha x[n] + \beta z[n]$	$\alpha\hat{X}(z) + \beta\hat{Z}(z)$	RoC(y) $\supseteq$ RoC(x_1) $\cap$ RoC(x_2)	Linearity
$x[n-m]$	$z^{-m}\hat{X}(z)$	RoC(x) $\setminus \{0\}$	Delay
$a^n x[n]$	$\hat{X}(z/a)$	aRoC(x)	Modulation
$\sum_{k=-\infty}^{\infty} x_1[k]x_2[n-k]$	$\hat{X}_1(z)\hat{X}_2(z)$	RoC(y) $\supseteq$ RoC(x_1) $\cap$ RoC(x_2)	Convolution
$x[n]^*$	$\hat{X}(z^*)^*$	RoC(y) = RoC(x)	Conjugation
$x[-n]$	$\hat{X}(z^{-1})$	RoC(y) = $\{z \in \mathbb{C} : z^{-1} \in \text{RoC}(x)\}$	Time-reversal

Table 2: Properties of the Z-transform. In the table, m is an integer, a, α and β are complex numbers.

Causality and BIBO Stability of DT LTI systems

Recall that we can determine important system properties from the impulse response, $h[n]$. Since the transfer function is simply the $\mathcal{Z}$-transform of $h[n]$, we can use its RoC and poles to study those same properties.

① Causality

A DT LTI system is causal if its impulse response satisfies:

$$h[n] = 0 \quad \text{for} \quad n < 0$$

In other words, its impulse response is right sided

For a causal DT LTI system with a rational transfer function, the RoC must therefore lie outside the outermost pole.

$$\text{RoC}(h): |z| > |p_{max}|$$

where $|p_{max}|$ is the largest pole magnitude

② BIBO stability

Recall from Week 3 that an ordinary DT LTI system is BIBO stable precisely when its impulse response is absolutely summable.

$$\sum_{n=-\infty}^{\infty} |h[n]| < \infty$$

But we have just learned that a sequence is absolutely summable precisely when its $\mathcal{Z}$-transform RoC contains the unit circle. Therefore,

$$\text{BIBO stable} \longleftrightarrow \text{RoC}(h) \text{ contains } |z| = 1$$

Combining causality & stability

Now we combine the 2 conditions:

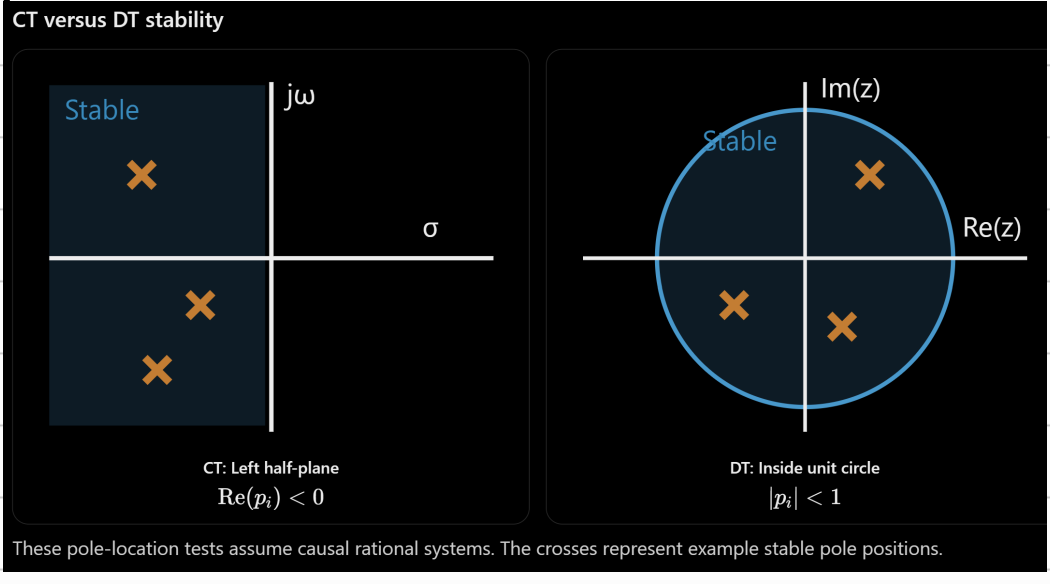
↳ Causal : RoC extend to the outside of the outermost pole

↳ BIBO : RoC must include unit circle ($|z| = 1$)

Therefore, to contain the unit circle, every pole must have magnitude less than 1

Causal rational DT LTI system is BIBO stable $\longleftrightarrow$ $|p_i| < 1$ for every pole p_i

Concept Check



For non-causal systems, poles outside unit circle do not automatically imply instability cuz non-causal system allow for the RoC to be the interior of the circle